JWST/NIRCam F090W and F150W Surface Brightness Fluctuation Distances from GO-3055: TRGB-Anchored Calibrations and Internal Accuracy Tests

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Abstract

We measure JWST/NIRCam F090W and F150W surface brightness fluctuation (SBF) distances for 14 early-type galaxies in program GO-3055 and calibrate both bands directly with the individual JWST tip of the red giant branch (TRGB) distances from Chazov et al. (2026). Every primary fit retains all 14 galaxies and uses galaxy-level leave-one-out (LOO) cross-validation. In F150W the data do not require a population correction: the adopted constant calibration is $\overline{M}_{150,0} = -3.208 \pm 0.024_{\rm stat} \pm 0.051_{\rm common}$ mag, with intrinsic scatter 0.081 mag and LOO RMS 0.0925 mag (4.3 per cent in distance). A linear color term worsens the LOO RMS to 0.0981 mag and is disfavored by $\Delta\mathrm{AICc} = 3.15$. In F090W, by contrast, color is informative. The adopted relation is $\overline{M}_{090,0} = -1.9617{+}1.8890[(F090W{-}F150W)_0{-}0.5703]$ mag, with intrinsic scatter 0.0489 mag and LOO RMS 0.0684 mag, compared with 0.0992 mag for a constant. The median full absolute distance uncertainty is 0.0752 mag for the F090W color calibration and 0.0984 mag for the F150W constant calibration; both include the common 0.047-mag TRGB scale. We report measurement, population, finite-calibration, and common-scale uncertainties separately and place all methods on the same error convention in the final distance table. These results are internal cross-validations of the calibrators, not independent external tests.
Keywords: distance scale; galaxies; surface brightness fluctuations; JWST; NIRCam; TRGB.

1 Introduction

Surface brightness fluctuations arise from Poisson variations in the number and luminosity of unresolved stars within each resolution element [1]. Their apparent amplitude decreases with distance, so an absolute SBF magnitude calibrated for the stellar population provides a distance indicator. In the near-infrared, evolved stars produce a strong SBF signal, but the absolute amplitude may depend on age and metallicity. Color is therefore commonly used as a population correction; it is an optional improvement to the distance estimator, not the definition of the method.

The TRGB–SBF project is constructing a Population-II distance ladder that transfers a geometrically calibrated TRGB zero point to SBF. Jensen et al. [Paper III; 13] used JWST TRGB distances to recalibrate the existing HST/WFC3-IR F110W SBF scale. With mean Virgo and Fornax distances, they obtained

$$\overline{M}_{110} = (1.86 \pm 0.16)[(g - z)_{\rm PS} - 1.30] - (2.760 \pm 0.016).$$

Here the SBF signal is measured independently in the JWST GO-3055 F090W and F150W images. The $F090W - F150W$ color never enters either power-spectrum measurement: it is introduced only after $\overline{m}$ has been measured, when the absolute fluctuation magnitude is calibrated. We test a color-independent and a color-dependent distance calibration in each band rather than assuming the same stellar-population response for F090W and F150W. The anchors are the revised JWST TRGB distances of Chazov et al. [Paper IV; 14], tied to the NGC 4258 maser distance. Our main questions are deliberately ordered as follows: (1) can the SBF amplitude be recovered consistently in both bands; (2) does color improve either distance calibration; (3) how accurately do the selected F150W and F090W relations recover the TRGB distances; and (4) are the propagated uncertainties consistent with the observed residuals?

2 Data and distance anchors

2.1 GO-3055 sample

The sample contains 14 galaxies: three Fornax members, nine galaxies in or near the Virgo region, and two nearby group galaxies. In tables and plotting legends, "Virgo" or "Virgo region" is only this broad sky/environment class; it is not a claim of strict Virgo Cluster membership. In particular, NGC 4636 and foreground NGC 4697 are excluded from the strict seven-object Virgo Cluster subset. Every target has NIRCam F090W and F150W imaging. The signal and color are measured in the same two circular annuli, 8.2–16.4 and 16.4–32.8 arcsec, following the geometry of Paper III. At the individual Paper IV distances, the outer 32.8-arcsec radius spans 1.65–3.17 kpc. Consequently, radial differences cannot be interpreted as measurements at a common physical radius. Jensen et al. also used an auxiliary elliptical-annulus check for elongated galaxies; that secondary test is not reproduced here, and the limitation is kept explicit.

The primary anchors are the individual Paper IV thin-slice TRGB moduli. This choice was fixed before inspection of the present SBF residuals and is kept to avoid selecting an anchor variant after seeing which gives less scatter. Paper IV prefers its thick-slice estimator because it uses a broader and more robust stellar sample on the blue side of the TRGB color break. We therefore treat thick-slice distances as a mandatory sensitivity experiment rather than silently ignoring that preference.

Two Jensen comparisons are kept separate throughout. Paper III is the HST F110W calibration of 14 TRGB–SBF targets and is used for the color-law and cluster-mean comparisons. A legacy Jensen et al. (2015) F160W reconstruction exists for only five common galaxies and is reported only as a separately labelled distance column; it is never merged with the Paper III F110W sample.

The extinction-corrected measurements, calibration-point uncertainties, and LOO residuals are reported together in Table 1 only after the measurement and statistical procedures have been defined. This keeps the data description separate from the derived results.

2.2 Foreground extinction

Paper IV tabulates A_{090}. Using its coefficients, we adopt

$$E(B-V) = A_{090}/1.4156, \tag{1}$$

$$A_{150} = 0.6021E(B-V), \tag{2}$$

$$C_0 = C_{\rm obs} - (A_{090} - A_{150}), \tag{3}$$

$$\overline{m}_{150,0} = \overline{m}_{150,\rm obs} - A_{150}. \tag{4}$$

The corrections make the colors bluer by 0.008–0.029 mag and the F150W SBF magnitudes brighter by 0.006–0.022 mag. They do not materially change the distance RMS, but they are required for an unbiased zero point and covariance matrix.

No separate internal-extinction correction is applied. The central 8.2 arcsec are excluded, obvious non-elliptical features enter the object mask, and the sample contains no retained strong dust lane. As a quantitative sensitivity bound, an unrecognized screen with $A_V^{\rm int} \leq 0.05$ mag would give $A_{150}^{\rm int} \lesssim 0.01$ mag for a standard near-infrared extinction curve, well below the observed 0.0925-mag LOO scatter. This is a bound, not a measurement of internal dust.

All targets have $z < 0.007$. For a deliberately broad local spectral-slope range $|1+\alpha| \leq 3$, where $f_\nu \propto \nu^\alpha$, the ordinary same-filter correction satisfies

$$|K_{150}| = 2.5|1+\alpha|\log_{10}(1+z) < 0.023 \text{ mag}. \tag{5}$$

It is not applied object by object because a consistent SBF spectral-energy model is unavailable and the bound is subdominant at the present precision. Likewise, cosmological surface-brightness dimming is not added as a separate SBF term: the common surface-brightness factor cancels in the variance divided by mean-galaxy-light normalization. Even the full Tolman term is bounded by $10\log_{10}(1+z) < 0.030$ mag over the sample. These two bounds delimit, rather than conceal, the small-redshift approximation.

3 Method overview

The same measurement chain is applied independently to F090W and F150W. For each image we subtract a scalar background, fit a smooth isophotal galaxy model, mask invalid pixels and compact sources, divide the residual by the square root of the model, and measure the PSF-shaped part of its Fourier power spectrum in two fixed annuli. The resulting apparent fluctuation magnitude $\overline{m}_X$ is a direct image measurement in band X. TRGB distances and color enter only in the subsequent calibration of the absolute magnitude $\overline{M}_X = \overline{m}_X - \mu_{\rm TRGB}$.

3.1 Sky, galaxy model, and masks

The input images are the public calibrated NIRCam i2d mosaics in MJy sr^{-1}. A scalar sky is estimated outside the science annuli from independent background regions and subtracted. Invalid pixels, detector gaps, foreground objects, and compact sources form the primary mask. Initial compact sources require at least nine connected pixels above a local 3.5σ threshold; after galaxy subtraction, a second catalogue uses four connected pixels above 2.5σ, deblending, and a three-pixel morphological dilation. The mask is deliberately aggressive toward compact positive residuals while retaining diffuse galaxy structure.

The centre is initialized from the image and checked against the WCS target position. Ellipticity and position angle remain free while isophotes are fitted from a starting semimajor axis of 50 pixels to at most 2000 pixels. Intensities and fitted geometry are linearly interpolated between consecutive isophotes; there is no piecewise constant filling between isophotal lines. A Sersic image is used only to fill detector gaps while solving the isophotes. The final model at every real science pixel comes from the measured isophotal profile. Filled gap pixels never enter a power spectrum.

After sky and galaxy-model subtraction, the residual is first divided by $\sqrt{M_i}$. The centre and scale of this *normalized* field are then estimated with up to five iterations of symmetric 3.5σ-clipped statistics, using every catalogue-unmasked pixel for which the science image is finite and $M_i > 0$. This support is the complete valid positive-model region: it is not the raw-residual distribution and not the union of the two measurement annuli. Values outside the resulting median plus or minus 3.5 scale interval are replaced by the corresponding boundary value. Pixels are not rejected. This operation is therefore *winsorization*, not sigma-clipping; sigma clipping is used only to estimate the centre and scale.

The resulting full-frame normalized and winsorized image is saved as a FITS product before any annulus is selected. Pixels rejected by the catalogue mask, invalid science pixels, and pixels with a non-positive model remain NaN in this product and never enter either the threshold estimate or a Fourier spectrum. The previous raw-residual and annulus-union prescriptions are retained only as comparison branches. Robustification is a deliberate implementation change relative to Jensen and is therefore tested explicitly rather than presented as part of the reproduced method.

Figure 1 shows the F150W product over the actual measurement support. Masked pixels are plotted in black; they are absent from both $P(k)$ and $E(k)$ rather than being assigned a numerical residual value. The saved FITS contains the full valid normalized and winsorized field. For the display, as for the subsequent FFT, the two annuli are selected and the mean of each annulus is removed. Thus mean subtraction belongs to construction of the Fourier input and is not baked into the reusable full-frame FITS.

4 SBF calculation

4.1 Normalized residual and expectation spectrum

For a smooth model M_i and sky-subtracted data I_i, define the normalized residual

$$Q_i = \frac{I_i - M_i}{\sqrt{M_i}}. \tag{6}$$

The division by $\sqrt{M_i}$ converts the Poisson stellar variance, which is proportional to local galaxy intensity, into an approximately stationary SBF amplitude. It does not measure a difference between the brightest and faintest pixels. SBF is the variance of the unresolved stellar luminosity function per unit mean light, recovered statistically from all usable pixels.

Let Q_i^{win} be Q_i after applying the single full-support winsorization interval described above, and let W_i be the binary window of one selected annulus. For that annulus the exact field entering the Fourier analysis is

$$R_i = W_i \left(Q_i^{\mathrm{win}} - \langle Q^{\mathrm{win}} \rangle_W\right). \tag{7}$$

Thus the threshold is estimated once over the full valid image, whereas the mean is estimated separately after selecting each ring. NaN pixels in the saved FITS are absent from the mean and from the FFT; zeros outside W_i in the explicit FFT array represent multiplication by the binary window, not measured zero-valued residuals. We calculate the two-dimensional FFT and use

$$P(\boldsymbol{k}) = \frac{|\mathcal{F}[R](\boldsymbol{k})|^2}{N_{\mathrm{use}}}. \tag{8}$$

The spectrum is azimuthally averaged into 80 radial bins; the uncertainty of each bin is the standard error of its contributing Fourier modes. The expected SBF spectrum is not simply $|\widetilde{\mathrm{PSF}}|^2$. For each of five time- and position-matched, unit-normalized STPSF models, 64 independent unit white-noise fields are convolved with the PSF, multiplied by the exact annular window and source mask, mean-subtracted, Fourier transformed, and averaged:

$$E_j(k) = \left\langle \frac{|\mathcal{F}\{W[n * \mathrm{PSF}_j]\}|^2}{N_{\mathrm{use}}} \right\rangle_{64}. \tag{9}$$

Thus $E(k)$ includes PSF blurring, the finite annulus, holes in the source mask, and mode mixing by the window.

The same N_{use}^{-1} Fourier normalization is applied to the measured field and every Monte Carlo realization used to construct $E(k)$, and every PSF has unit summed flux before convolution. Therefore the fitted coefficient P_0 is already on the fluctuation-flux scale defined by Eq. 7. No additional division by the mean model brightness is permitted after the fit; doing so would normalize the galaxy light twice. In the diagnostic power-spectrum panels, the black points are the azimuthal means of the measured two-dimensional power spectrum in radial Fourier bins, and their error bars are the standard errors of the contributing Fourier modes. They are not individual image pixels or PSF samples.

4.2 Power-spectrum fit and fluctuation magnitude

For every PSF realization we fit

$$P(k) = P_{0,j} E_j(k) + P_{1,j} \tag{10}$$

by weighted least squares, with weights equal to the inverse squared bin errors. The covariance is $(X^{\mathsf{T}} W X)^{-1}$ multiplied by $\max(\chi^2_\nu, 1)$. The adopted P_0 and P_1 are the medians over the five PSFs. The primary normalized-frequency interval is $0.04 \le k \le 0.25$, with an additional lower bound of ten Fourier modes across the shorter crop axis. Fits beginning at $k_{\min} = 0.01$ and 0.03, with the same $k_{\max}$, quantify low-frequency sensitivity but do not replace the primary fit.

The PSF-shaped power contains both stellar SBF and unresolved compact sources:

$$P_{\mathrm{SBF}} = P_0 - P_r. \tag{11}$$

Here P_r is specifically the contribution from unmasked globular clusters and background galaxies fainter than the catalogue limit. In each annulus the observed source luminosity function is modelled as a Gaussian globular-cluster luminosity function plus a power-law background-galaxy term. Its faint side is integrated with flux-squared weighting,

$$P_r = \langle M^{-1} \rangle \int_{m_{\mathrm{cut}}}^{m_{\mathrm{cut}}+12} [\phi_{\mathrm{GC}}(m) + \phi_{\mathrm{bg}}(m)] f^2(m)\, dm, \tag{12}$$

where $f(m)$ is the flux in image units and $\langle M^{-1} \rangle$ restores the normalization of Eq. 7. The adopted GCLF width is 1.2 mag; widths of 1.35 mag for NGC 1399 and NGC 4472 are tested separately.

Because Eq. 7 already divides the variance by mean galaxy light, P_{SBF} has units of fluctuation flux and is not divided by M a second time. With pixel solid angle Ω_{pix} and band X,

$$Z_{\mathrm{AB,pix},X} = -2.5 \log_{10}\left(\frac{10^6 \Omega_{\mathrm{pix}}}{3631}\right), \tag{13}$$

$$\overline{m}_X = -2.5 \log_{10}(P_0 - P_r) + Z_{\mathrm{AB,pix},X}. \tag{14}$$

The inner and outer magnitudes are finally combined with inverse-variance weights. Their half-difference is

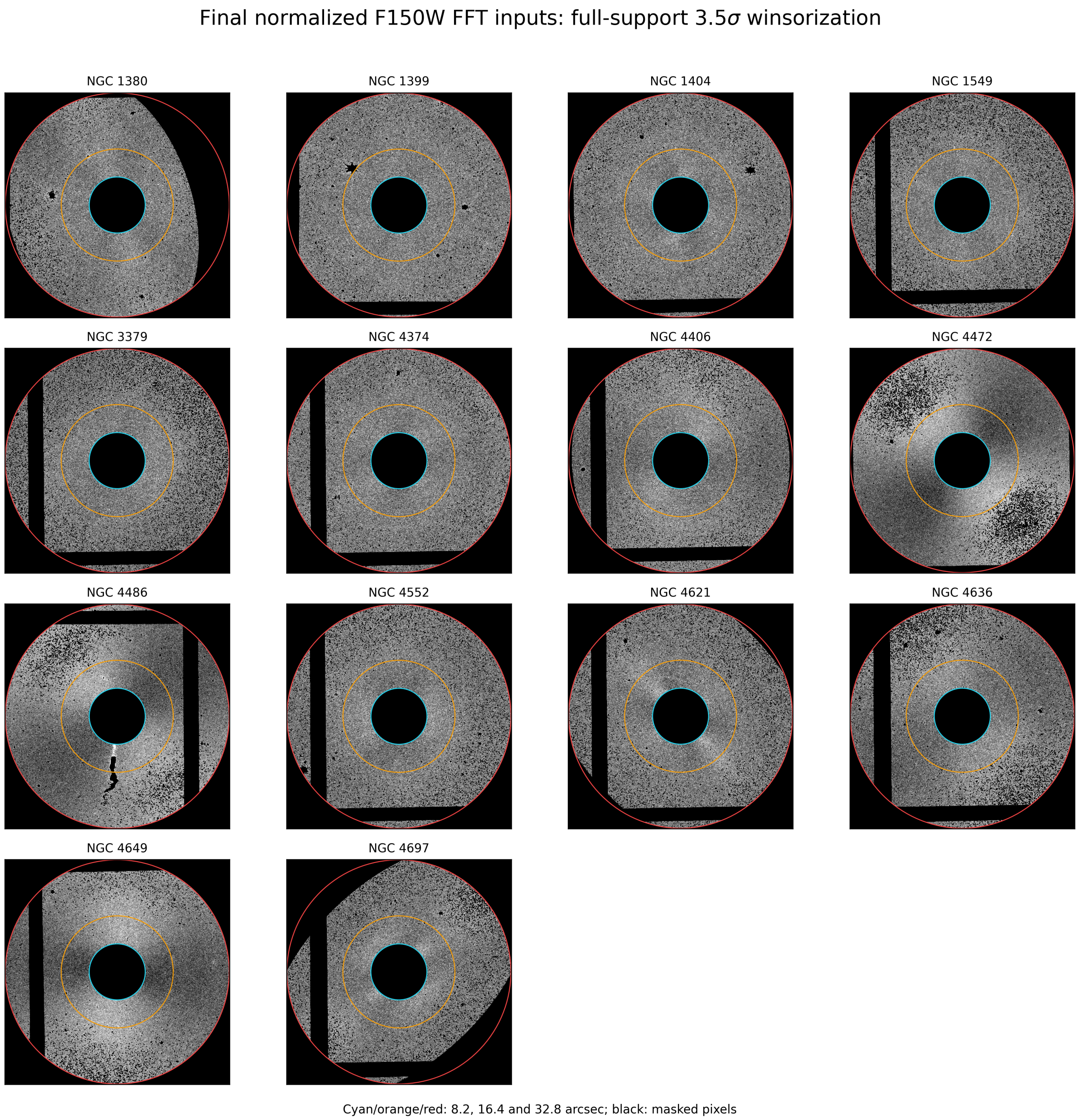

Figure 1: Final F150W Fourier inputs constructed from the full-frame residual products after division by $\sqrt{M_i}$ and symmetric 3.5σ winsorization. The smooth galaxy model has been subtracted; compact sources, invalid image regions, and pixels outside the two measurement annuli are black. Visible grey-scale structure is therefore an unmasked residual, not a mask value. The cyan, orange, and red circles mark 8.2, 16.4, and 32.8 arcsec. The displayed value in each annulus has had that annulus mean removed.

retained as a radial diagnostic, not inserted as random measurement noise.

All current annular fits give $P_1 < 0$. Corner regions of the original F150W i2d frames have a median logarithmic noise-spectrum slope of -1.45 over the fitted frequency range, so the data are not white. The fitted constant absorbs the local intercept of this misspecified noise spectrum and may be negative; it is not a claim of negative physical noise power.

4.3 Matched color and calibration variable

F090W is registered and PSF-matched to F150W. Color is measured in the same usable pixels and rings as the SBF signal. The two annular colors may differ because the stellar population changes with radius. Their half-difference is therefore retained as a radial diagnostic, not assigned as a statistical color error. Color does not enter either power-spectrum measurement. It is tested only at the calibration stage: F150W adopts a constant $\overline{M}_{150}$, whereas the F090W results below adopt a linear $\overline{M}_{090}$–color relation because it improves both AICc and LOO prediction.

5 Uncertainty model

We keep four different quantities separate: the uncertainty of the measured apparent fluctuation magnitude $\overline{m}_X$, the uncertainty of one calibration point $\overline{M}_X = \overline{m}_X - \mu_{\mathrm{TRGB}}$, the predictive uncertainty of a new SBF distance, and the common uncertainty of the absolute distance scale. Confusing these levels is the reason that a 0.02–0.03-mag image measurement can coexist with a 0.07–0.10-mag distance uncertainty. Common terms move all galaxies coherently and are never used as independent regression weights.

5.1 Ring-level spectrum and apparent SBF error

Let r denote the inner or outer annulus. The non-PSF power-spectrum error is defined conservatively as

$$\sigma_{k,r} = 1.4826\,\mathrm{MAD}\{\overline{m}_r(k_{\min})\}, \tag{15}$$

$$\sigma_{\mathrm{WLS},r} = \frac{1.085736\,\sigma_{P_0,\mathrm{WLS}}}{P_0 - P_r}, \tag{16}$$

$$\sigma_{\mathrm{PS},r} = \max(\sigma_{k,r}, \sigma_{\mathrm{WLS},r}). \tag{17}$$

where the MAD uses $k_{\min} = 0.01, 0.03$, and 0.04 and the second term is the linear propagation of the WLS covariance. It is a maximum, not a quadrature sum, because both terms diagnose uncertainty in the same fitted amplitude. The factor $1.085736 = 2.5/\ln 10$ converts fractional power error to magnitude. This duplicate-estimator maximum is our conservative implementation choice, not a separate physical noise source and not an exact prescription quoted from Jensen et al.

The five PSFs are then kept separate. Their robust scatter is

$$\sigma_{\mathrm{PSF},r} = 1.4826\,\mathrm{MAD}\{\overline{m}_{r,j}\}. \tag{18}$$

The analysis notebook reconstructs Eq. 17 from the saved $\sigma_{P_0,\mathrm{WLS}}$ before adding this term; hence PSF scatter is not counted both inside and outside the fit error. With the same normalized inverse-variance annular weights w_r used for the central magnitude,

$$w_r = \frac{\sigma_r^{-2}}{\sum_s \sigma_s^{-2}}, \qquad \overline{m}_X = \sum_r w_r \overline{m}_{X,r}, \tag{19}$$

$$\sigma_{\mathrm{PS}}^2 = \sum_r w_r^2 \sigma_{\mathrm{PS},r}^2, \tag{20}$$

$$\sigma_{\mathrm{PSF}} = \sum_r w_r \sigma_{\mathrm{PSF},r}. \tag{21}$$

Here σ_r is the saved ring-level spectral uncertainty; the weights are not proportional to annulus area. Independent spectral errors combine in quadrature, while the PSF terms are combined linearly because a single PSF mismatch moves both annuli coherently.

The scalar-sky systematic s is the scatter among independent background estimates in image units. Linearizing Eq. 14 gives

$$\sigma_{\mathrm{sky},r} = 1.085736\frac{s}{\langle M\rangle_r}, \qquad \sigma_{\mathrm{sky}} = \sum_r w_r \sigma_{\mathrm{sky},r}. \tag{22}$$

Again the annuli are correlated because the same scalar sky is subtracted from both. If $q_r = P_r/P_0$, assigning the stated 25 per cent fractional uncertainty to P_r gives

$$\sigma_{P_r,r} = 1.085736\,(0.25)\frac{q_r}{1-q_r}, \qquad \sigma_{P_r} = \sum_r w_r \sigma_{P_r,r}. \tag{23}$$

This term is small because $P_r/P_0 < 0.0067$ for every retained annulus; the 0–$2P_r$ sensitivity experiment tests the stronger possibility that the working luminosity-function model itself is wrong.

The internal apparent-SBF error is therefore

$$\sigma_{\overline{m},\mathrm{int}}^2 = \sigma_{\mathrm{PS}}^2 + \sigma_{\mathrm{PSF}}^2 + \sigma_{\mathrm{sky}}^2 + \sigma_{P_r}^2. \tag{24}$$

The radial half-difference does not appear in Eq. 24: it can be a real population gradient and the two rings share their sky, model, mask, and PSF.

5.2 Extinction and calibration-point errors

For foreground reddening we assign

$$\sigma_{A_{090}}^2 = (0.16A_{090})^2 + 0.005^2, \tag{25}$$

$$\sigma_{E(B-V)} = \sigma_{A_{090}}/1.4156, \tag{26}$$

$$\sigma_{E(090-150)} = (1.4156 - 0.6021)\sigma_{E(B-V)}. \tag{27}$$

The individual Paper IV modulus error already contains its A_{090} term. We remove that contribution, $\sigma_{\mu,\mathrm{no\ red}}^2 = \sigma_{\mu,\mathrm{IV}}^2 - \sigma_{A_{090}}^2$, and then restore the single correct shared-reddening combination. For an F150W absolute calibration point this leaves the differential extinction $A_{090} - A_{150}$:

$$\sigma^2_{\overline{M}_{150},i} = \sigma^2_{\overline{m},\mathrm{int}} + \sigma^2_{\mu,\mathrm{no\ red}} + \sigma^2_{E(090-150)}. \quad (28)$$

The same reddening moves corrected color and absolute SBF in opposite directions, giving

$$\mathrm{Cov}(C_0, \overline{M}_{150,0}) = -\sigma^2_{E(090-150)}. \quad (29)$$

For F090W, the same A_{090} occurs in the corrected SBF magnitude and in the Paper IV TRGB modulus. It therefore cancels from $\overline{M}_{090} = (\overline{m}_{090} - A_{090}) - \mu_{\mathrm{TRGB}}$, and

$$\sigma^2_{\overline{M}_{090},i} = \sigma^2_{\overline{m}_{090},\mathrm{int}} + \sigma^2_{\mu,\mathrm{no\ red}}, \qquad \mathrm{Cov}(C_0, \overline{M}_{090,0}) = 0. \quad (30)$$

This cancellation is specific to calibrating F090W against a TRGB modulus whose foreground correction is also determined in F090W; it is not an assumption that extinction itself vanishes.

The photometric color error in each ring is the standard error of the usable pixel-color distribution after WCS registration and PSF matching. The two ring errors are combined as $\sigma^2_{C,\mathrm{phot}} = \sum_r w_r^2 \sigma^2_{C,r}$. The horizontal error used in the diagnostic linear regression is $\sigma^2_C = \sigma^2_{C,\mathrm{phot}} + \sigma^2_{E(090-150)}$. This is a photometric error; the observed inner–outer color difference remains a separate radial-gradient diagnostic. The standard error does not claim to include every image-wide F090W/F150W calibration or PSF-matching systematic. Adding a deliberately larger 0.01-mag independent color floor is retained as a sensitivity test. It does not change the F150W conclusion and does not remove the F090W preference for a color-dependent calibration.

5.3 Calibration likelihood and predictive distance errors

The adopted F150W calibration is color independent,

$$\overline{M}_{150,0} = a, \qquad \mu_{\mathrm{SBF}} = \overline{m}_{150,0} - a. \quad (31)$$

Consequently, neither the measured color nor its uncertainty enters an F150W distance. In F090W the adopted $f(C)$ is linear and its color term does enter the distance prediction, but never the preceding FFT measurement.

For a color law $f(C)$, the likelihood uses the effective variance

$$V_i = \sigma^2_{\overline{M},i} + [f'(C_i)\sigma_{C,i}]^2 - 2f'(C_i)\mathrm{Cov}(C_i, \overline{M}_i) + \sigma^2_{\mathrm{int}}. \quad (32)$$

and maximizes the Gaussian likelihood including $\ln V_i$. For the adopted F150W constant relation, $f'(C) = 0$, so all color and color-covariance terms vanish. For F090W, Eq. 30 gives zero reddening covariance of the calibration point, while the horizontal color uncertainty remains in V_i. The intrinsic term σ_{int} is therefore fitted, not chosen to force reduced $\chi^2 = 1$. Fits with no horizontal error, an additional 0.01-mag object-to-object floor, and the historical half-ring proxy are reported as sensitivity tests.

For a target in band X, differentiating $\mu_{\mathrm{SBF}} = \overline{m}_{X,\mathrm{obs}} - R_X E(B-V) - f_X(C_0)$ gives

$$\sigma^2_{\mathrm{target},X} = \sigma^2_{\overline{m}_X,\mathrm{int}} + [f'_X(C)\sigma_{C,\mathrm{phot}}]^2 \quad (33)$$
$$+ [R_X - f'_X(C)(R_{090} - R_{150})]^2 \sigma^2_{E(B-V)}. \quad (34)$$

This propagates the covariance between extinction-corrected color and SBF magnitude once, rather than adding correlated terms independently. It reduces to $\sigma^2_{\overline{m}_{150},\mathrm{int}} + \sigma^2_{A_{150}}$ for constant F150W. For linear F090W it contains $[R_{090} - b(R_{090} - R_{150})]^2 \sigma^2_E$. The internal predictive variance then separates target measurement, population scatter, and finite calibration:

$$\sigma^2_{\mu,\mathrm{int}} = \sigma^2_{\mathrm{target}} + \sigma^2_{\mathrm{int}} + \sigma^2_{\mathrm{cal}}. \quad (35)$$

The intrinsic term is the value fitted to the relevant 13-object LOO training sample. The finite-calibration term is the standard deviation of 300 non-parametric bootstrap predictions from those same 13 galaxies. The full absolute distance error is

$$\sigma^2_{\mu,\mathrm{abs}} = \sigma^2_{\mu,\mathrm{int}} + 0.047^2. \quad (36)$$

The common Paper IV TRGB/NGC 4258 zero point of 0.047 mag moves the whole distance scale coherently; it is not used as 14 independent regression errors. A common F150W photometric zero point and a common finite-PSF-stamp shift cancel to first order between an apparent F150W SBF magnitude and an F150W absolute calibration measured with the same pipeline, so they are not added again as independent distance errors. They do affect the quoted absolute $\overline{M}_{150}$ zero point: the 0.047 mag TRGB, 0.0108 mag flux-scale, and 0.0167 mag PSF-stamp terms combine to 0.0510 mag. The common TRGB term does not enter the relative calibration fit.

Explicitly, a one per cent common F150W flux-scale uncertainty becomes $\sigma_{\mathrm{flux}} = (2.5/\ln 10)(0.01) = 0.0108$ mag. The PSF-stamp term is the largest measured 129-versus-257-pixel spectral shift, 0.0167 mag. Therefore the common absolute-magnitude scale is

$$\sigma_{\overline{M},\mathrm{common}} = \sqrt{0.047^2 + 0.0108^2 + 0.0167^2} = 0.0510 \text{ mag}. \quad (37)$$

For a same-filter SBF distance, the last two terms shift apparent and calibrated absolute F150W magnitudes together and cancel; only the 0.047-mag anchor is retained. Any modulus uncertainty is converted to a fractional distance uncertainty with

$$\frac{\sigma_D}{D} = \frac{\ln 10}{5}\sigma_\mu, \quad (38)$$

using the linear form for the quoted percentages. The same propagation is used in the printed and machine-readable distance tables.

For a plotted validation residual $\Delta\mu = \mu_{\mathrm{SBF}} - \mu_{\mathrm{TRGB}}$, the common TRGB zero point cancels. The reddening response of that difference is

$$[R_{090} - R_X + f'_X(C)(R_{090} - R_{150})]\sigma_E. \quad (39)$$

It becomes $(R_{090} - R_{150})\sigma_E$ for the adopted constant F150W calibration and $b(R_{090} - R_{150})\sigma_E$ for the adopted F090W linear calibration. The validation variance combines this term with the SBF measurement, color photometry, individual TRGB error without reddening, intrinsic scatter, and finite-calibration error. It is therefore neither the raw fit error nor the full absolute-scale error. LOO uses the same calibrator set and is an internal cross-validation, not an external test on unseen anchors.

Jensen et al. (2021) also combine independent contributions in quadrature. In their representative F110W budget the median P_0-fit, sky, PSF, P_r, and extinction terms give $\sigma_{\overline{m}_{110}} \simeq 0.057$ mag; color and a 0.060-mag population term give $\sigma_{\overline{M}_{110}} \simeq 0.075$ mag, and the typical distance-modulus uncertainty is 0.085 mag. For the Paper III values in our final table we propagate the tabulated $\overline{m}_{110}$ and color errors, the published slope uncertainty, and the same 0.060-mag population scatter, then add the Paper III 0.063-mag common absolute scale. The published 0.016-mag TRGB-to-SBF zero-point term is already contained in that 0.063 mag and is not added twice. Paper IV TRGB and both of our JWST columns likewise include their common 0.047-mag scale in the final absolute-error table. These common terms remain correlated even though they are printed inside each distance cell.

The numerical contribution of each term is given with the corresponding passband results. Parentheses in compact summaries give the full minimum–maximum range across galaxies rather than an uncertainty on the median.

6 Calibration strategy and pre-specified tests

The constant and linear calibrations are compared with the same Gaussian likelihood, fitted intrinsic scatter, AICc, and galaxy-level LOO prediction. The LOO experiment removes the target before fitting its calibration and is therefore an internal cross-validation of the 14 calibrators, not an external validation on independent galaxies.

Both passbands use all 14 galaxies, individual Paper IV thin-slice distances, and no manual rejection. Constant and linear relations are compared before a distance estimator is selected. The F150W data select one color-independent zero point; the F090W data select the linear color relation. Quadratic, logarithmic, exponential, and higher-order relations remain exploratory even when one has a marginally smaller AICc. NGC 1399 remains in the primary fit; its constant-zero-point LOO residual is -0.143 mag, or 1.68σ with the validation uncertainty, and does not justify post-hoc removal.

The reported sensitivity experiments include Paper IV thick-slice and Paper-III-style cluster anchors; fixed quality subsets; inner and outer annuli; $P_r = 0$, the adopted P_r, and $2P_r$; GCLF-width variants; PSF stamp size and detector position; the order and support used for residual winsorization; and an optional correlated-noise template. Nonlinear color laws and removal of individual galaxies cannot replace the selected primary relations.

7 F150W results

7.1 Extinction-corrected measurements

Table 1 is the single object-level audit table for the primary experiment. It joins the measured color and apparent SBF magnitude to the individual TRGB anchor and the resulting LOO residual; it is not an additional fit or a quality-selected subsample.

7.2 Color dependence: not required by the F150W data

We first ask whether color is needed before constructing the primary distance estimator. The color-independent fit gives

$$\begin{aligned}\overline{M}_{150,0} &= -3.208 \pm 0.024_{\text{stat}} \pm 0.051_{\text{common}} \text{ mag}, \\ \sigma_{\text{int}} &= 0.081 \text{ mag}.\end{aligned} \tag{40}$$

The statistical term is the standard deviation of 2000 galaxy-level bootstrap zero points. The common term combines the TRGB/NGC 4258 scale, the adopted F150W flux-scale allowance, and the finite-PSF-stamp term; only the TRGB term remains in a same-filter SBF distance. The bootstrap 16th–84th percentile interval of σ_{int} is 0.064–0.088 mag. At the median corrected color $C_0 = 0.5675$, the alternative linear fit is

$$\overline{M}_{150,0} = -3.2094 + 0.2244(C_{090-150,0} - 0.5675), \tag{41}$$

with intrinsic scatter 0.0798 mag. Bootstrap 16th–84th percentiles are $[-3.2313, -3.1815]$ mag for the intercept and $[-0.3044, 0.7136]$ for the slope, so the slope interval includes zero.

The constant model is better than the linear relation by ΔAICc = 3.15 and also has lower LOO RMS. Color therefore does not improve the distance estimate in the present sample. This conclusion is not caused by the treatment of horizontal errors: the linear LOO RMS is 0.09809 mag with no color errors and 0.09810 mag either with the measured errors or after adding a 0.01-mag floor; using the old half-ring proxy gives 0.09847 mag. We consequently adopt Eq. 40; all subsequent primary distances and error budgets are color independent.

Quadratic and fixed-break broken-line relations were also tested only as exploratory shape checks. Their AICc/LOO-RMS pairs are $-19.77/0.0945$ and $-20.32/0.0922$ mag, respectively. The broken-line RMS is only 0.0003 mag below the constant-model value while its AICc is worse by 3.43. Neither model improves both criteria; their additional coefficients are therefore not promoted to a calibration table or a physical law.

The cleanest comparison with Paper III uses the same eight galaxies rather than comparing two different samples. For our F150W measurements versus $F090W - F150W$, the constant and linear models have LOO RMS values of 0.1014 and 0.1104 mag, respectively; the corresponding AICc values are -9.65 and -4.63. For the Paper III F110W measurements of those same galaxies versus $g - z$, the linear model reduces the LOO RMS from 0.1114 to 0.0465 mag and has AICc -11.94, compared with -7.97 for a constant. Thus the discrepancy is not produced merely by changing the galaxy list.

Table 1: Extinction-corrected GO-3055 measurements. The penultimate column is the color-independent LOO SBF-minus-TRGB residual. The printed "Virgo region" label denotes the broad environment class defined in Section 2.1. HQ IV reproduces the high-quality flag in Table 2 of Paper IV. The quoted $\sigma_{\overline{M}}$ contains individual errors but excludes common zero points. The annular half-difference is a radial diagnostic and is not included as random noise.

Galaxy	Environment	$C_{090-150,0}$ (mag)	$\overline{m}_{150,0}$ (mag)	$\overline{M}_{150,0}$ (mag)	$\sigma_{\overline{M}}$ (mag)	$\lvert\Delta\overline{m}\rvert_{\rm ann}$ (mag)	$\mu_{\rm TRGB}$ (mag)	$\Delta\mu_{\rm LOO,const}$ (mag)	HQ IV
NGC 1380	Fornax	0.510	28.087	-3.321	0.029	0.099	31.408	-0.122	+
NGC 1399	Fornax	0.570	28.157	-3.341	0.034	0.047	31.498	-0.143	−
NGC 1404	Fornax	0.559	28.177	-3.189	0.028	0.110	31.366	+0.020	+
NGC 1549	Other	0.533	27.930	-3.301	0.029	0.133	31.231	-0.101	−
NGC 3379	Other	0.557	27.012	-3.066	0.030	0.269	30.078	+0.152	+
NGC 4374	Virgo region	0.566	27.947	-3.165	0.031	0.094	31.112	+0.046	+
NGC 4406	Virgo region	0.616	27.849	-3.247	0.030	0.097	31.096	-0.042	+
NGC 4472	Virgo region	0.625	27.713	-3.303	0.035	0.029	31.016	-0.103	−
NGC 4486	Virgo region	0.652	27.808	-3.238	0.055	0.054	31.046	-0.033	−
NGC 4552	Virgo region	0.569	27.850	-3.073	0.030	0.175	30.923	+0.146	+
NGC 4621	Virgo region	0.565	27.688	-3.133	0.028	0.173	30.821	+0.081	+
NGC 4636	Virgo region	0.604	27.890	-3.180	0.031	0.110	31.070	+0.031	+
NGC 4649	Virgo region	0.623	27.873	-3.136	0.030	0.043	31.009	+0.077	−
NGC 4697	Virgo region	0.512	27.095	-3.229	0.032	0.173	30.324	-0.023	+

Table 2: Primary model comparison for all 14 galaxies. The linear relation tests whether a population correction improves on the adopted constant.

Model	a	b	$\sigma_{\rm int}$	AICc / LOO RMS
Constant	-3.208	0.000	0.081	-23.75 / 0.0925
Linear	-3.209	0.224	0.080	-20.60 / 0.0981

A crossed test of both SBF bands against both available colors constrains, but does not uniquely identify, the source of the discrepancy. On the same eight galaxies, F150W has Pearson $r = 0.28$ ($p = 0.504$) against $F090W - F150W$ and $r = 0.21$ ($p = 0.611$) against $g - z$. F110W gives $r = 0.82$ ($p = 0.0125$) and $r = 0.93$ ($p = 0.0010$), respectively. The two color indices themselves remain strongly correlated. Thus the F110W relation survives either color choice whereas the F150W relation does not. This points to the SBF passband as an important difference, but the eight objects still mix passband response, adopted distance prescription, environment, and reduction choices; the test does not isolate a unique physical cause. It also does not show that color is intrinsically useless or that no F150W color dependence can exist in a broader sample.

7.3 Distance recovery and precision

The adopted color-independent relation gives LOO bias -0.0010 mag and RMS 0.0925 mag, corresponding to approximately 4.3 per cent scatter in distance. Its median predicted internal uncertainty is 0.0865 mag (4.1 per cent), and the median absolute uncertainty after the common zero point is 0.0984 mag (4.6 per cent).

The constant-zero-point validation pulls have RMS 1.08; eight of 14 objects lie within 1σ and all 14 within 2σ. Thus the error model is not obviously underestimated or inflated. The empirical LOO RMS and the median predicted uncertainty are distinct quantities and agree at the expected level for this small sample.

The adopted F150W LOO distances are combined with F090W, TRGB, and the two distinct Jensen comparisons only in the final distance table immediately before the Discussion. Keeping that table out of the band-specific section prevents formal errors from one method being mistaken for full predictive errors from another.

7.4 Error distribution by galaxy

Individual calibration-point errors are typically dominated by the TRGB anchor and power-spectrum measurement. NGC 4486 is exceptional because its background term is 0.0456 mag. In the predictive distance budget, intrinsic scatter is the largest component, followed by finite-calibration uncertainty; improving the formal P_0 error alone cannot reduce the total distance error below the present 4–5 per cent floor.

Table 3 summarizes the galaxy-dependent terms; Table 4 propagates them through an actual LOO prediction; and Table 5 lists coherent scale errors that must not be added as though they were independent for each galaxy. The largest median calibration-point component is the individual TRGB error. The exceptional 0.0456-mag sky term belongs to NGC 4486.

7.5 Measurement-pipeline systematics

The local STPSF library contains five unit-normalized models per galaxy. Four offsets around the nominal detector position change the fitted SBF magnitude by at most 0.0043 mag. Stamp size matters more: relative to the 129-pixel model, truncation to 97, 65, and 33 pixels can shift the fitted amplitude by as much as 0.0127, 0.0433, and 0.1248 mag, respectively. More importantly, independently generated 257-pixel STPSFs show that the central 129 pixels miss 0.69–0.77 per cent of the 257-pixel

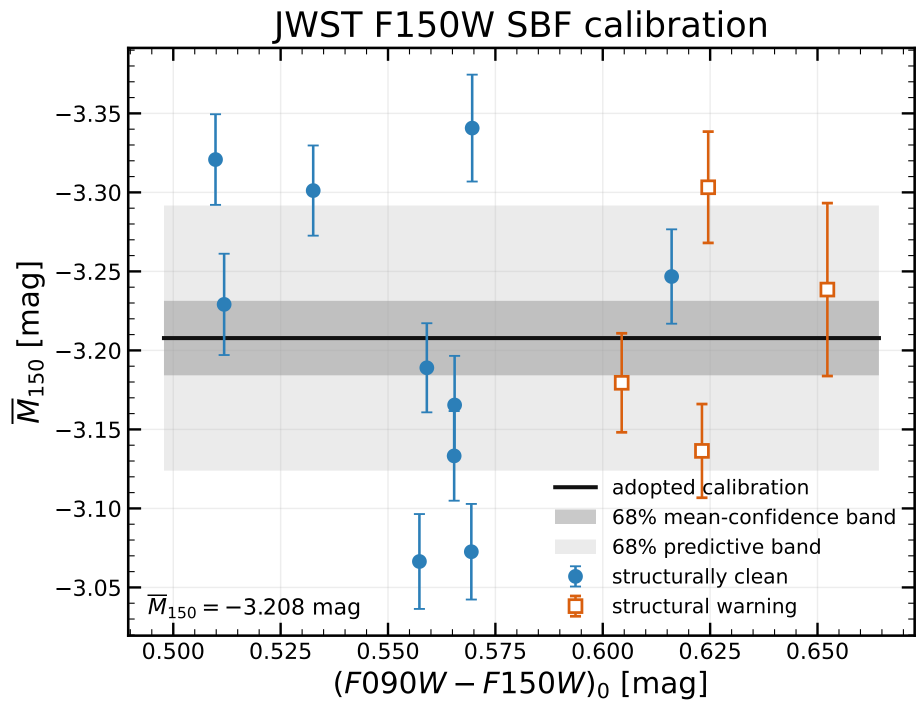

Figure 2: Extinction-corrected JWST F150W SBF calibration. The black horizontal line is the adopted color-independent zero point. The dark grey region is the central 68 per cent confidence interval for the mean calibration; the lighter region is the 68 per cent predictive interval after inclusion of the fitted intrinsic scatter. Filled blue circles form the structurally clean subsample; open orange squares mark the predeclared structural-warning objects. All 14 objects enter the adopted calibration.

Table 3: Galaxy-dependent statistical uncertainties in magnitudes. Values are the median and, in parentheses, the minimum–maximum range across the 14 calibration points. Intrinsic and finite-calibration terms apply to predictions rather than to a single measured calibration point.

Component	Median	Range
Power-spectrum fit and k-window	0.0131	0.0096–0.0222
STPSF realization scatter	0.0022	0.0012–0.0033
Residual sources P_r	0.0006	0.0000–0.0013
Sky subtraction	0.0071	0.0035–0.0456
Paper IV TRGB (without reddening)	0.0251	0.0238–0.0271
Shared foreground reddening	0.0040	0.0031–0.0054
Total individual calibration point	0.0301	0.0282–0.0547

flux and imply a common spectral shift of 0.0151–0.0167 mag over the fitted k range. We retain 0.0167 mag as a common absolute-$\overline{M}_{150}$ systematic. Including it with the TRGB and NIRCam terms gives a 0.0510-mag absolute zero-point uncertainty; it does not change the internal LOO residuals.

The production size is 129 pixels, not 127. The odd dimension provides an unambiguous central sample and 64 pixels on either side. It retains 99.23–99.31 per cent of the corresponding 257-pixel STPSF flux for the three tested targets while reducing the repeated Monte Carlo convolution cost. The 257-pixel experiment shows that this choice produces a nearly coherent 0.015–0.017-mag absolute shift, rather than an uncontrolled object-to-object scatter. Retaining 129 pixels is therefore justified for the present same-pipeline distance comparison provided that the measured 0.0167-mag term is carried on the absolute $\overline{M}_{150}$ scale; a future sub-percent absolute calibration should instead use the larger stamp.

An empirical check was attempted with six bright, isolated compact sources in the NGC 3379 i2d frame. Their median stack is broader than STPSF and gives a formal 0.288-mag spectral difference. This number is *not* adopted as a PSF correction: the objects are not confirmed single stars, the i2d product has no DQ extension, and residual host/background structure broadens the stack. Thus the model-internal finite-stamp behavior is constrained, but an absolute empirical PSF zero point remains open until an isolated classified star is available. The detailed PSF diagnostics are shown in Appendix A; they are numerical validation plots, not additional astrophysical measurements.

The adopted 3.5σ residual winsorization was checked for all 14 galaxies with the galaxy model, catalogue mask, annular windows, PSFs, $E(k)$, and P_r held fixed. Four spectral branches were compared: no winsorization; winsorization of the raw residual before normalization; winsorization after normalization with a threshold estimated over the complete valid positive-model support; and the same operation with a threshold estimated only from the union of the two annuli. The third prescription is adopted because its threshold is defined in the same normalized variable that is Fourier analysed, while still producing a reusable full-frame FITS before the annuli are selected.

Relative to the former raw-residual prescription, the adopted full-support branch changes the weighted $\overline{m}_{150}$

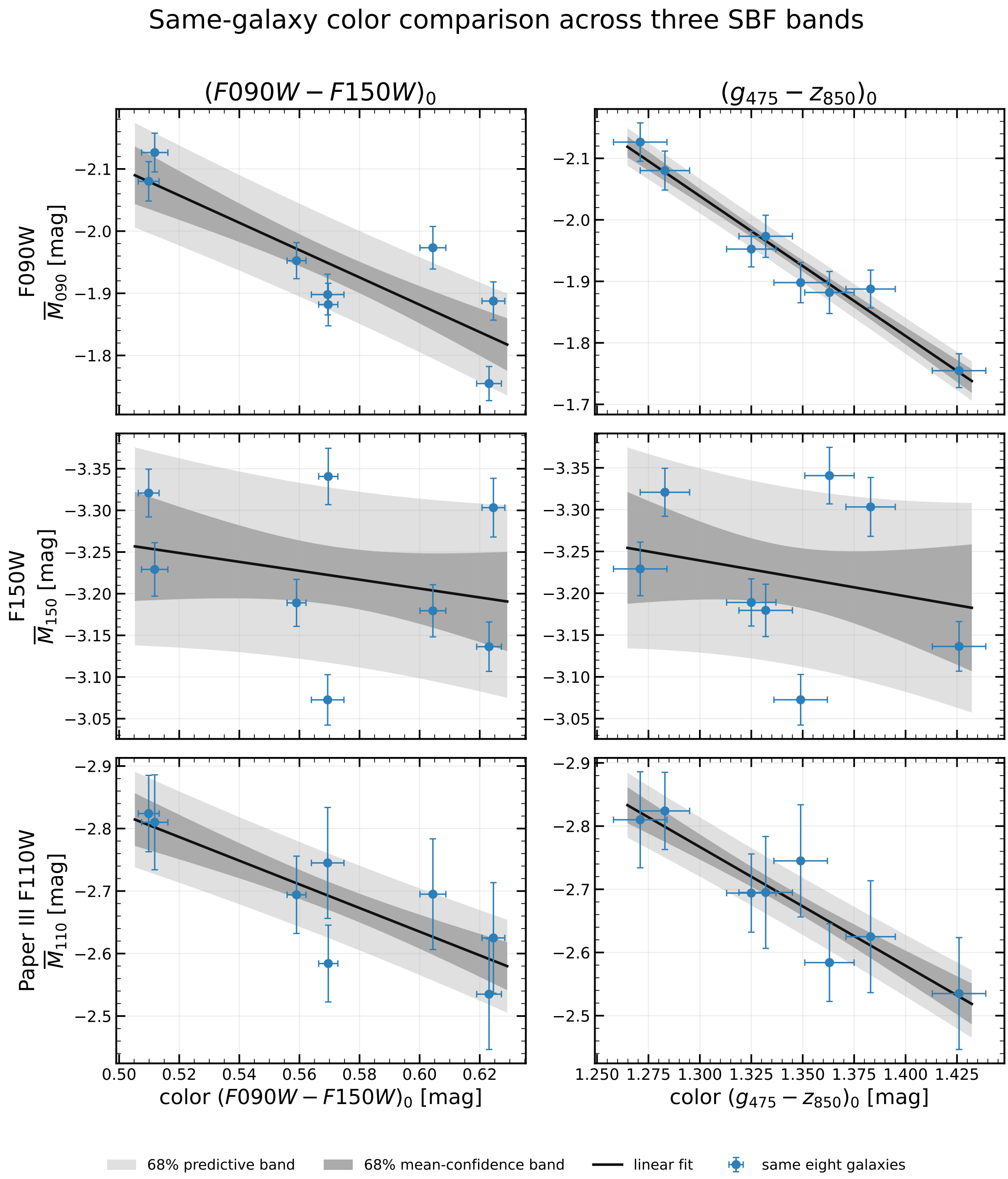

Figure 3: Same-galaxy color test for the eight objects common to this work and Paper III. Rows show F090W, F150W, and Paper III F110W; columns use $F090W - F150W$ and $g - z$. Thus neither the galaxy list nor the color tracer is changed in isolation. F090W and F110W retain clear slopes, whereas F150W does not. In every panel the dark grey region is the central 68 per cent confidence interval for the mean linear relation and the lighter region is the 68 per cent predictive interval including residual scatter.

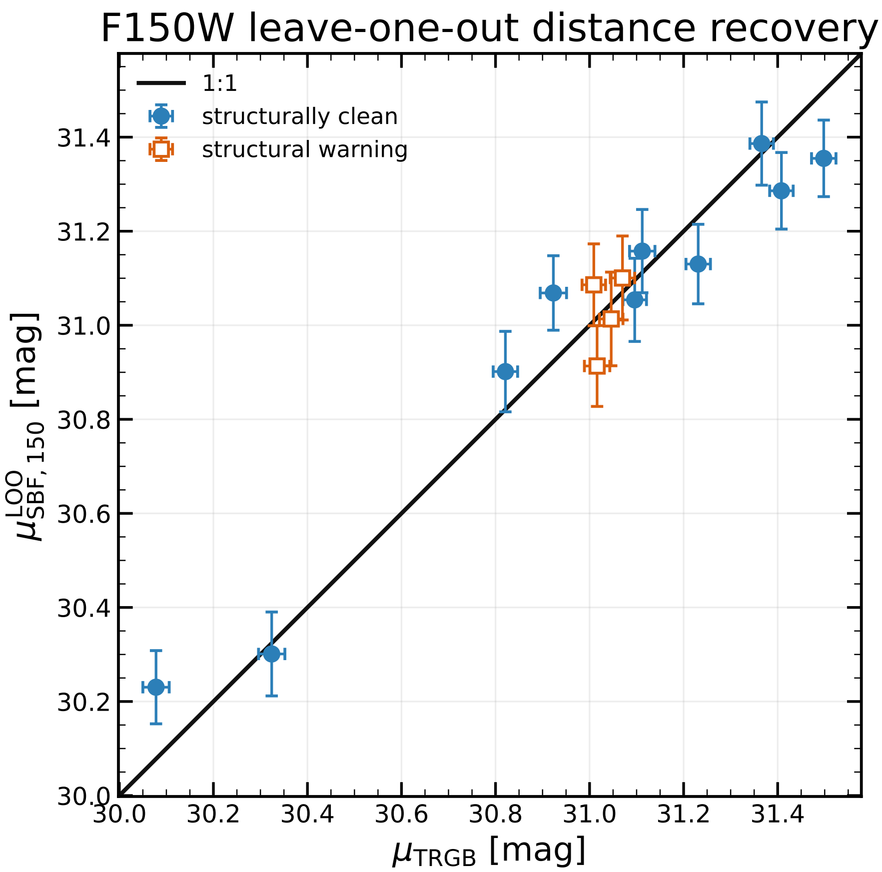

Figure 4: Color-independent LOO F150W SBF distance moduli compared with the individual Paper IV thin-slice TRGB moduli. Each target is excluded from its own calibration. The dashed line is equality. Error bars are internal; the common absolute zero point is not shown.

Table 4: Decomposition of a predicted distance-modulus uncertainty. Entries are median (minimum–maximum) in magnitudes across the 14 LOO targets. Only the adopted color-independent calibration is shown. "Internal" combines target measurement, extinction, intrinsic scatter, and finite calibration. "Absolute" adds the common 0.047-mag TRGB scale. The final validation row also includes the individual target TRGB error and is not an uncertainty for a new SBF-only distance. Linear-model terms remain available only in the machine-readable sensitivity table.

Component	Adopted constant
Apparent SBF measurement	0.0169 (0.0122–0.0490)
Foreground-extinction propagation	0.0030 (0.0023–0.0040)
Fitted intrinsic population scatter	0.0814 (0.0734–0.0839)
Finite 13-galaxy LOO calibration	0.0240 (0.0215–0.0262)
Total internal SBF distance	0.0865 (0.0778–0.0996)
Total including common TRGB scale	0.0984 (0.0909–0.1101)
SBF–TRGB validation residual	0.0901 (0.0825–0.1025)

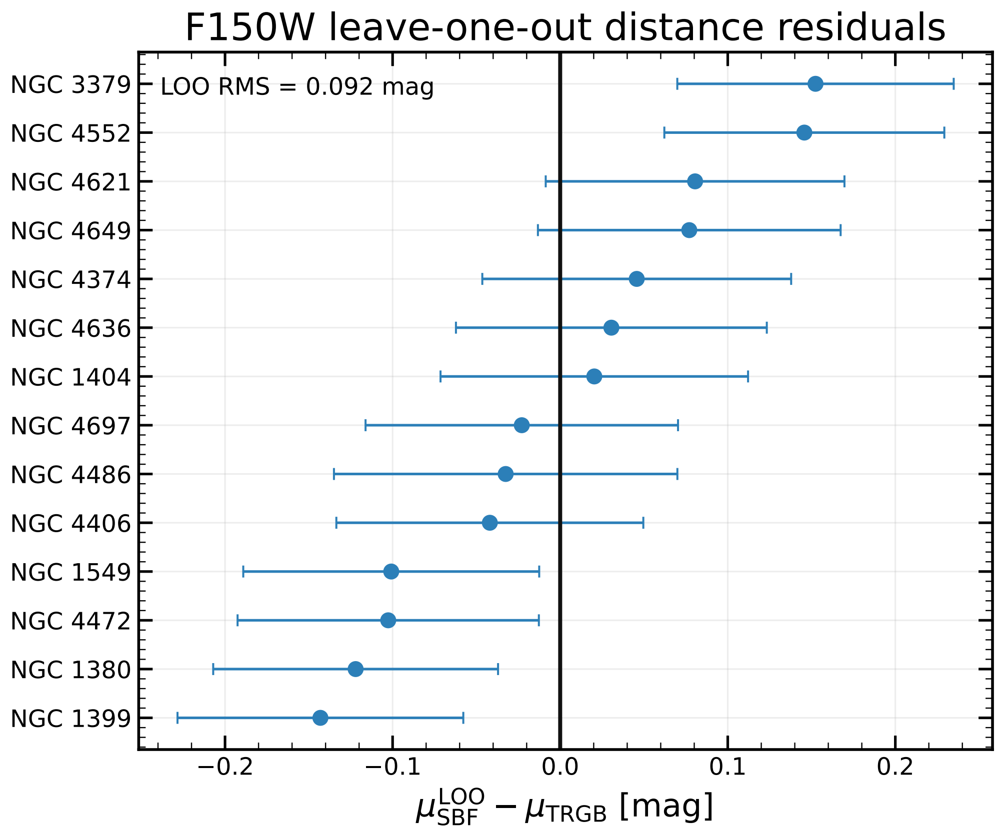

Figure 5: Residuals of the color-independent F150W LOO predictions relative to the individual Paper IV thin-slice TRGB moduli. The vertical dashed line marks zero residual. Error bars contain the galaxy-dependent predictive terms but exclude the shared absolute TRGB zero-point scale.

Table 5: Common systematic uncertainties. The combined absolute $\overline{M}_{150}$ scale is the quadrature of the preceding calibration terms. The same-filter distance row is a separate cancellation case and is not added to that total. The 1 per cent F150W flux-scale entry is a conservative common analysis allowance, not an uncertainty measured from these 14 galaxies.

Component	σ (mag)	Use
TRGB/NGC 4258 zero point	0.0470	absolute calibration
NIRCam F150W flux scale	0.0108	absolute calibration
STPSF finite-stamp term	0.0167	absolute calibration
Combined absolute $\overline{M}_{150}$ scale	0.0510	absolute calibration
Same-filter SBF distance scale	0.0470	F150W SBF distances

by a median of −0.02954 mag over the 14 galaxies. The largest absolute change is 0.08798 mag for NGC 4649. The ordering of these operations is consequently part of the final measurement definition, not a negligible perturbation and not an extra random error added after the fact. This winsorization remains an explicit departure from the Jensen pipeline; all alternative branches are retained as a sensitivity experiment rather than mixed into the primary calibration.

An extended model $P(k) = P_0 E(k) + P_n N(k)$ was tested using an empirical correlated-noise spectrum. Over the available frequency interval, the $E(k)$–$N(k)$ template correlations are 0.982–0.997. The additional coefficient therefore makes P_0 unstable and, in some fits, unphysical. We retain the Jensen-style $P_0 E(k) + P_1$ model. A negative P_1 is allowed as the local intercept of a misspecified correlated-noise continuum; it is not interpreted as negative physical noise power.

The unresolved-source correction is negligible at the present precision. Across all galaxies, replacing the adopted P_r by either zero or $2P_r$ changes an individual weighted SBF magnitude by at most 0.0053 mag within a fixed residual measurement. This is much smaller than the 0.0925-mag cross-validated scatter of the adopted calibration. Re-fitting NGC 1399 and NGC 4472 with the Jensen-specific GCLF width of 1.35 mag changes any weighted magnitude by at most 0.00046 mag. Artificial-source completeness remains desirable, but it cannot alter the present calibration at a scientifically relevant level unless the current P_r estimate is wrong by far more than a factor of two.

A synthetic FFT-stage test passes a known normalized SBF field, mask, PSF, and white-noise term through the FFT and $P_0 E(k) + P_1$ fit. Over 32 realizations the recovered magnitude has mean bias 0.0078 ± 0.0067 mag (standard error of the mean), consistent with zero; the trial-to-trial RMS is 0.0382 mag. The test validates the core unit conversion and spectral normalization, but not isophotal modelling, source detection, or the correlated i2d noise. The P_r and FFT-stage diagnostic plots are also collected in Appendix A.

A separate branch-recovery experiment isolates the order of normalization and winsorization. Sixty-four independent synthetic fields per annulus use a known

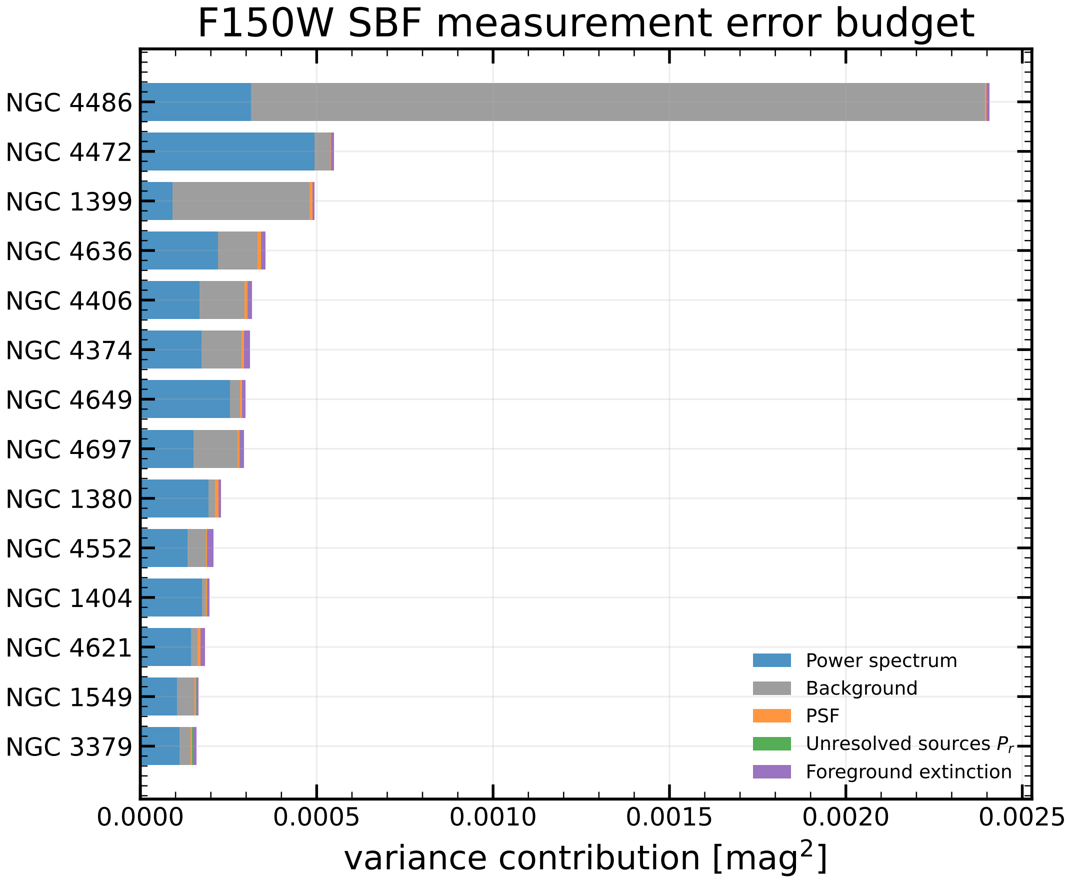

Figure 6: F150W apparent-SBF measurement budget in the coursework-style variance representation. Each horizontal segment is one contribution σ_i^2; the total measurement uncertainty is the square root of the complete bar length. Galaxies are ordered by total measurement variance.

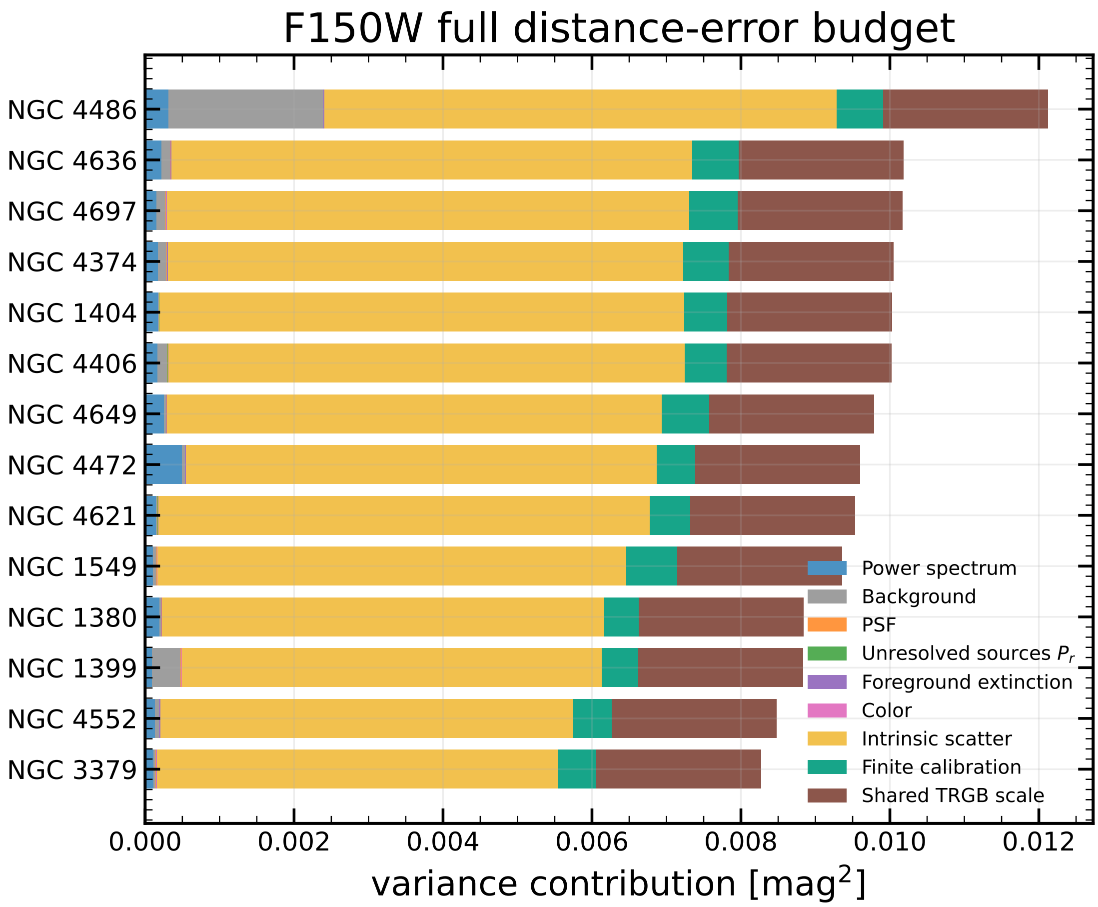

Figure 7: Full F150W LOO distance-error budget in the same variance representation. In addition to the image-measurement terms, the bars include intrinsic calibration scatter, finite-sample calibration uncertainty, and the shared TRGB absolute scale. The latter is coherent between galaxies and must not be interpreted as 14 independent random contributions.

P_0, a radially varying positive galaxy model, the saved 129-pixel PSF, catalogue-like holes, and the same four spectral branches. Relative to the no-winsorization result for the same realization, the adopted full-support branch shifts the recovered magnitude by 0.00104 ± 0.00008 mag in the inner annulus and 0.00098 ± 0.00005 mag in the outer annulus (mean and standard error). Thus the chosen ordering does not introduce a detectable additional bias for an injected Gaussian SBF field. This is deliberately not described as a full end-to-end validation: it does not reproduce sky subtraction, isophotal modelling, source detection, unresolved-source completeness, or the non-Gaussian residual structure of the real mosaics.

7.6 Distance-anchor sensitivity

Table 6 compares the pre-specified thin-slice analysis, the Paper IV preferred thick-slice values, and a Paper-III-style experiment. The latter assigns the published Paper III means $\mu_{\mathrm{Fornax}} = 31.424 \pm 0.057$ mag and $\mu_{\mathrm{Virgo}} = 31.055 \pm 0.086$ mag to cluster members; NGC 1549, NGC 3379, and foreground NGC 4697 retain individual anchors.

For the adopted constant calibration, the thin/thick choice changes the zero point by only 0.006 mag and the LOO RMS by less than 0.001 mag. Cluster means reduce the fitted intrinsic scatter, as expected when depth and individual TRGB noise are suppressed, yet the gain against individual thin-slice distances is small. No conclusion depends on selecting the thin variant over the Paper IV preferred thick estimator. For the thin, thick, and cluster-mean anchors, the diagnostic linear-model LOO RMS values against their own anchors are 0.0981, 0.0994, and 0.0961 mag, respectively; none improves on the corresponding constant result.

7.7 Independent-band comparison with legacy F160W SBF

Five galaxies overlap the independently reduced Jensen et al. (2015) F160W sample. Their weighted mean modulus difference is $\mu_{150} - \mu_{160} = -0.117 \pm 0.070$ mag and the weighted RMS is 0.107 mag. The rank correlation between the two absolute SBF measurements is suggestive (Spearman $\rho = 0.87$, $p = 0.054$), but the ordinary Pearson test is not significant ($r = 0.65$, $p = 0.233$). A fluctuation-color slope is likewise not detected. With only five common galaxies, this is a useful cross-band consistency check and a possible small offset to monitor, not an independent validation of the F150W distance scale. The object-level comparison is shown in Appendix B.

7.8 Annular behavior and environment

The inner annuli give slope 0.98 and intrinsic scatter 0.051 mag; the outer annuli give slope 0.07 and scatter 0.091 mag. The two points per galaxy are correlated and are not 28 independent calibrators. NGC 3379 has an exceptionally uniform residual image but a 0.269 mag annular SBF difference, demonstrating why the annular difference cannot be treated automatically as random measurement noise.

For the eight galaxies shared with Paper III, the direct F150W–F110W comparison is also substantially more coherent in the inner annulus (Pearson $r = 0.74$, $p = 0.035$; slope 0.56 ± 0.21) than in the outer annulus ($r = 0.31$, $p = 0.448$; slope 0.29 ± 0.36). In this population-sensitive sense, the inner measurements are more stable: they show the clearer cross-filter ordering and lower inter-galaxy scatter. This is not a blanket statement that the inner ring is numerically more robust. The all-14 spectral branch comparison gives the opposite result for sensitivity to high-amplitude residuals: the outer rings change less when the normalization and winsorization prescription is varied. The two notions of "stability" must not be conflated, and the present results do not support an order-of-magnitude precision claim for either ring.

Both circular rings are retained and combined because reproducing the radial geometry of Jensen/Paper III is a primary design constraint. They do not represent a claim that fixed angular annuli sample identical physical populations, nor that the outer ring is intrinsically preferable. A future optimization may choose physical or elliptical apertures, but that would be a new calibration experiment rather than a literal replication.

For the strict Virgo/Fornax subset, the seven Virgo members are NGC 4374, NGC 4406, NGC 4472, NGC 4486, NGC 4552, NGC 4621, and NGC 4649; the three Fornax members are NGC 1380, NGC 1399, and NGC 1404. A fitted Fornax shift is -0.157 mag. However, an exact permutation test gives $p = 0.150$, color and cluster membership are strongly confounded, and the step model is worse by $\Delta\mathrm{AICc} = 0.59$. The step is a diagnostic, not a detection. Foreground extinction does not remove this pattern.

7.9 Structural-quality sensitivity

Quality flags were declared independently of the calibration fit. NGC 4486 contains the jet and strong large-scale structure; NGC 4472, NGC 4636, and NGC 4649 have strong residual structure; and NGC 1380 and NGC 4697 have incomplete outer model coverage. The primary sample nevertheless remains 14/14. Restricting it to the 11 visually acceptable or 10 automatically clean objects does not improve cross-validation: the constant-model LOO RMS becomes 0.0982 and 0.1036 mag, respectively, compared with 0.0925 mag for all 14. The nine Paper IV high-quality objects give 0.0869 mag; their linear model is worse at 0.1037 mag and therefore does not restore a useful color term. The test supports retaining the full sample while reporting the flags rather than using image appearance for post-hoc rejection.

8 F090W results

8.1 Extinction-corrected measurements

Table 7 reports the F090W measurement before any color law is selected. The apparent magnitude $\overline{m}_{090,0}$ comes

Table 6: Sensitivity of the adopted constant calibration to the TRGB distance prescription. The final column always evaluates LOO predictions against the primary individual Paper IV thin-slice distances. Cluster-mean LOO values are not an independent galaxy-by-galaxy validation because different galaxies share an anchor.

Anchor	a	$\sigma_{\rm int}$	LOO RMS vs anchor	LOO RMS vs thin
Paper IV thin	-3.208	0.081	0.0925	0.0925
Paper IV thick	-3.202	0.081	0.0932	0.0927
Paper III cluster means	-3.231	0.070	0.0924	0.0949

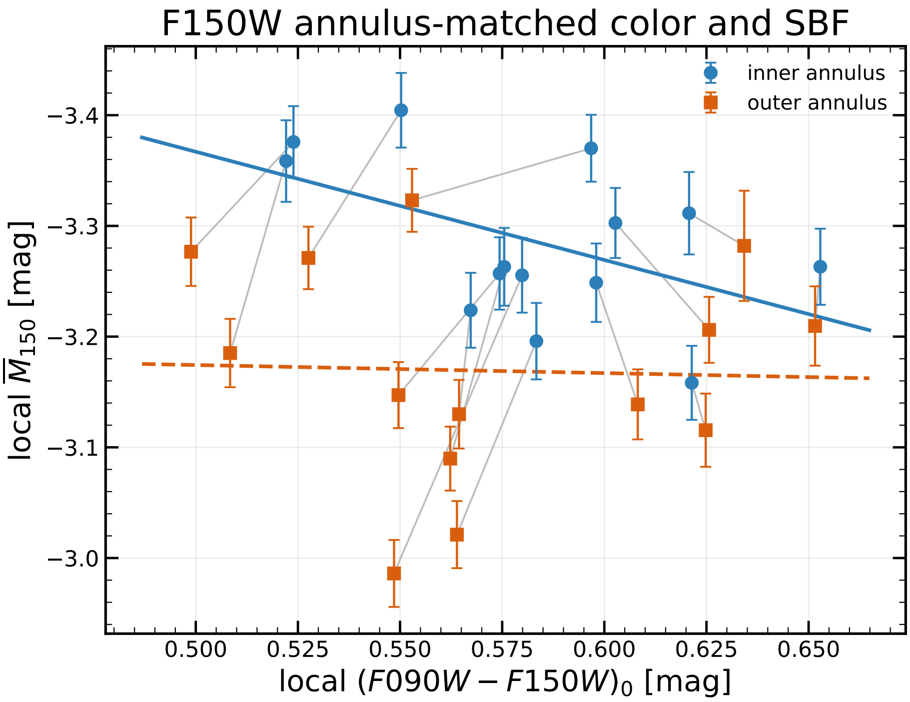

Figure 8: Local color and absolute F150W SBF magnitude in matched annuli. Blue circles and the solid blue line denote inner-annulus measurements and their fit; orange squares and the dashed orange line denote the outer annulus. Grey segments connect the two measurements of each galaxy. No uncertainty corridors are added: the figure is a direct radial comparison, and neither annular fit defines the adopted calibration.

from the same two-ring FFT procedure as F150W. The tabulated $\overline{M}_{090,0}$ uses the individual Paper IV distance only to display the calibration point; it is not the output of a fitted color relation.

Figure 9 is the direct F090W counterpart of Figure 1. It uses the same masks, annular geometry, normalization, winsorization rule, mean removal, and display convention, so visual differences between the two montages trace the data rather than a change of plotting recipe.

8.2 Color dependence and adopted calibration

Unlike F150W, F090W requires a population correction over the color range of this sample. With $C_0 = (F090W - F150W)_0$ and pivot $C_p = 0.5703$ mag, the adopted full-sample relation is

$$\overline{M}_{090,0} = (-1.9617 \pm 0.0157_{\mathrm{boot}}) + (1.8890 \pm 0.4424_{\mathrm{boot}})(C_0 - C_p), \tag{42}$$

with fitted intrinsic scatter 0.0489 mag. Relative to a constant, the linear model improves AICc by 9.15 and lowers the LOO RMS from 0.0992 to 0.0684 mag, a 31 per cent reduction. The logarithmic form has AICc lower by only 0.33 and LOO RMS 0.0673 mag. It is statistically tied with the linear relation and does not justify replacing the simpler, directly interpretable slope.

The color preference is strongest for the full prespecified sample. In the 11 visually clean, 10 automatically clean, and nine Paper IV high-quality subsets, the linear LOO RMS values are 0.0677, 0.0745, and 0.0656 mag, respectively. Their smaller sample sizes weaken formal model selection but do not reverse the fitted positive slope. These subset results remain sensitivity checks, not alternative calibrations.

8.3 Distance recovery and uncertainty budget

The adopted F090W linear relation has LOO bias -0.0039 mag and LOO RMS 0.0684 mag, equivalent to approximately 3.2 per cent distance scatter. Its median predictive uncertainty is 0.0587 mag internally and 0.0752 mag after adding the common TRGB scale; the latter corresponds to a median 0.57 Mpc for this sample. The validation-pull RMS is 1.12: nine objects lie within 1σ, 13 within 2σ, and NGC 4649 is the single 2.49σ case. Thus F090W is more precise than the color-independent F150W calibration on these calibrators, although one object warrants continued systematics scrutiny.

The median linear-model contributions are 0.0079 mag from the non-PSF power spectrum, 0.0147 mag from PSF realization, 0.0068 mag from the sky, and 0.00025 mag from P_r. The fitted intrinsic term is approximately 0.049 mag in the LOO samples and the median finite-calibration term is 0.0258 mag. The median propagated color-photometry and extinction terms are only 7.0×10^{-5} and 7.0×10^{-4} mag, respectively; the gain from color comes from removing a population trend, not from assigning a large color error.

The 129-versus-257-pixel F090W PSF experiment changes the fitted spectrum by at most 0.0128 mag and is retained as a coherent same-band diagnostic rather than added independently to every distance. Inner and outer F090W color-law slopes are 2.62 and 1.18, so radial behavior is not identical even though the combined relation is significant. For NGC 4636 the compact-source premask was relaxed only inside 8.2 arcsec while solving the isophotes; the final science mask and both measured annuli were not relaxed.

9 Adopted distances

All columns in Table 10 use one deliberately uniform convention: the printed uncertainty is the full absolute 1σ error in Mpc. Table 9 states which individual and common terms enter each method. These total errors are appropriate for an absolute distance quoted in isolation. They are not independent in a method-to-method residual, because shared TRGB and other common terms must then be cancelled or represented through a covariance matrix.

After this correction, the median full errors are 0.70 Mpc for the eight Paper III F110W overlaps, 0.75 Mpc for our F150W distances, and 0.57 Mpc for our F090W distances. The earlier apparent factor-of-two disadvantage of the JWST results was an error-convention mismatch: a formal Jensen measurement error had been compared with our full predictive error.

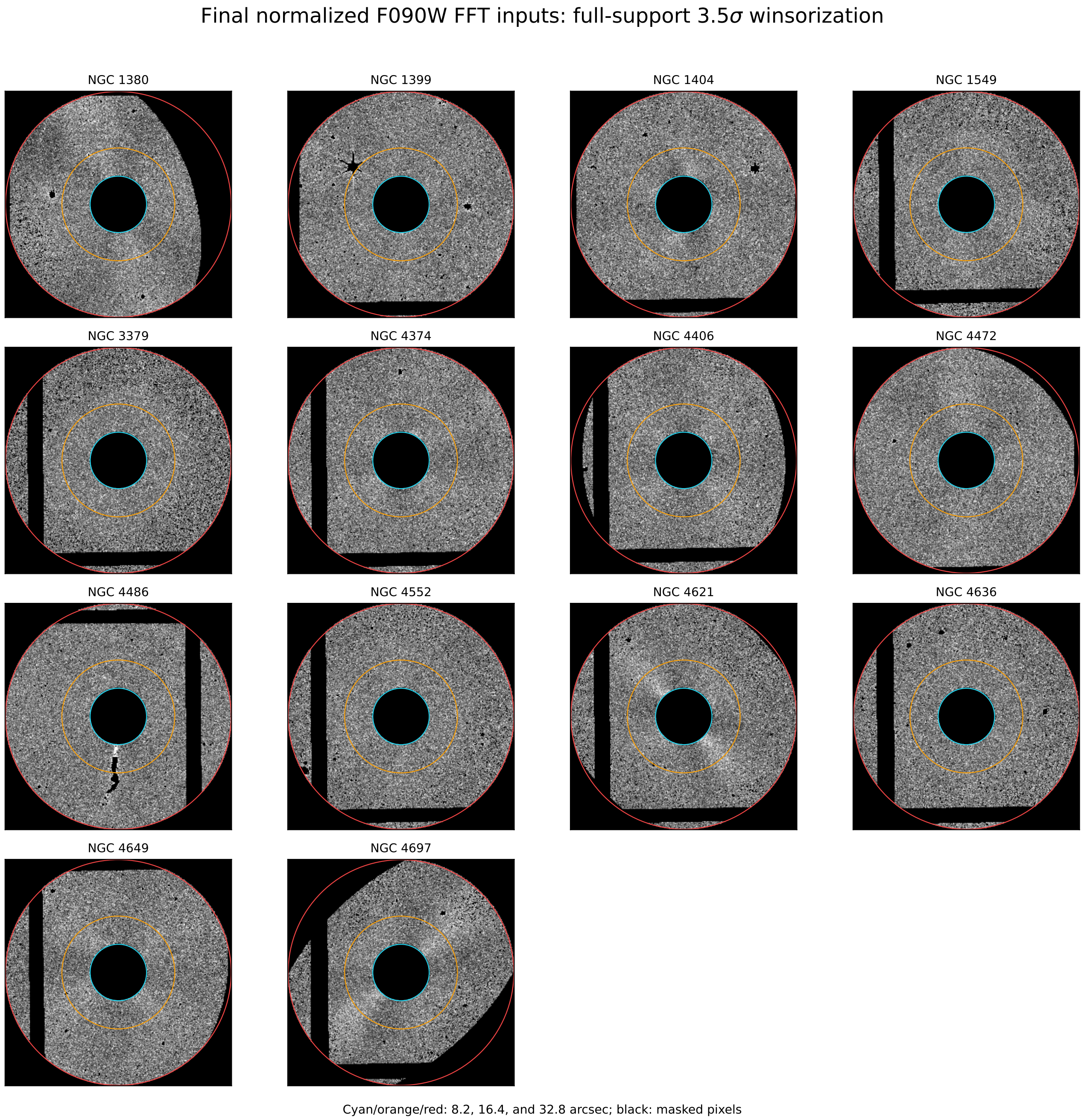

Figure 9: Final F090W Fourier inputs constructed from the full-frame residual products after division by $\sqrt{M_i}$ and symmetric 3.5σ winsorization. The plotting convention is identical to Figure 1: black denotes excluded pixels, the cyan, orange, and red circles mark 8.2, 16.4, and 32.8 arcsec, and the displayed value in each annulus has had that annulus mean removed.

Table 7: Extinction-corrected F090W SBF measurements. The quoted $\sigma_{\overline{m}}$ is the internal image-measurement uncertainty from Eq. 24; $\sigma_{\overline{M}}$ additionally contains the individual TRGB uncertainty according to Eq. 30 but excludes common zero points. The annular half-difference is shown only as a radial diagnostic.

Galaxy	Env.	$C_{090-150,0}$ (mag)	$\overline{m}_{090,0}$ (mag)	$\sigma_{\overline{m}}$ (mag)	$\overline{M}_{090,0}$ (mag)	$\sigma_{\overline{M}}$ (mag)	$\lvert\Delta\overline{m}\rvert_{\mathrm{ann}}$ (mag)	HQ IV
NGC 1380	Fornax	0.511	29.328	0.020	-2.080	0.032	0.039	+
NGC 1399	Fornax	0.569	29.616	0.023	-1.882	0.034	0.079	−
NGC 1404	Fornax	0.565	29.414	0.016	-1.952	0.029	0.006	+
NGC 1549	Other	0.539	29.214	0.027	-2.017	0.037	0.045	−
NGC 3379	Other	0.570	28.156	0.017	-1.922	0.032	0.156	+
NGC 4374	Virgo region	0.566	29.163	0.023	-1.949	0.034	0.029	+
NGC 4406	Virgo region	0.615	29.130	0.024	-1.966	0.034	0.040	+
NGC 4472	Virgo region	0.626	29.128	0.016	-1.888	0.031	0.018	−
NGC 4486	Virgo region	0.652	29.199	0.057	-1.847	0.062	0.056	−
NGC 4552	Virgo region	0.574	29.025	0.019	-1.898	0.033	0.088	+
NGC 4621	Virgo region	0.571	28.806	0.019	-2.015	0.031	0.075	+
NGC 4636	Virgo region	0.603	29.097	0.024	-1.973	0.034	0.041	+
NGC 4649	Virgo region	0.622	29.254	0.013	-1.755	0.027	0.045	−
NGC 4697	Virgo region	0.517	28.198	0.016	-2.126	0.031	0.057	+

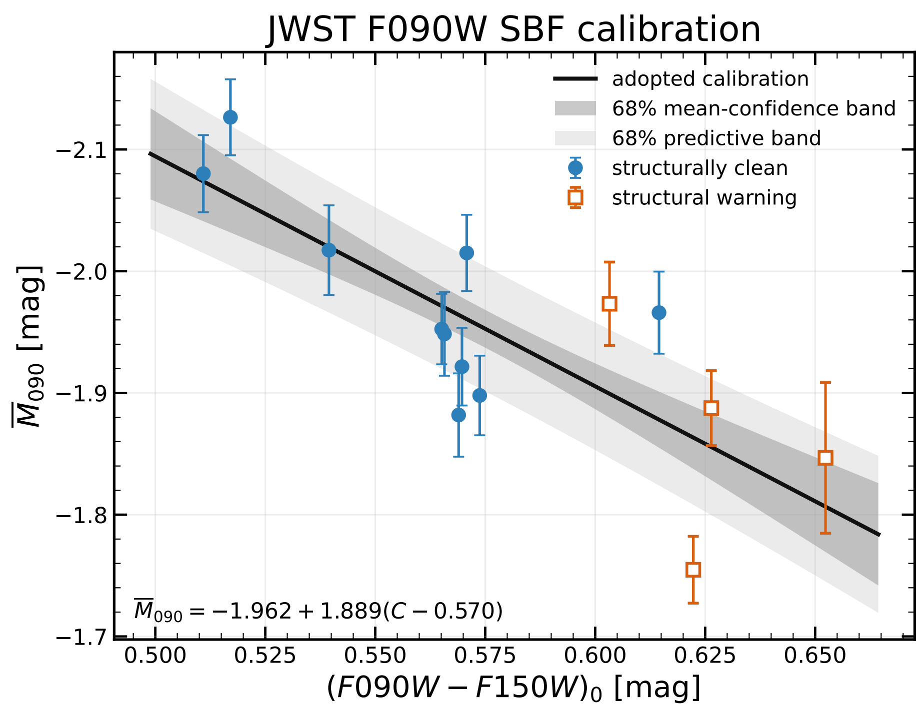

Figure 10: Adopted F090W color calibration. Points are the 14 individual $\overline{M}_{090}$ calibration measurements and the black line is Eq. 42. The dark grey region is the central 68 per cent confidence interval for the mean relation; the lighter region is the 68 per cent predictive interval after inclusion of the fitted intrinsic scatter. Filled blue circles and open orange squares have the same structural quality meaning as in Figure 2; all 14 objects enter the fit.

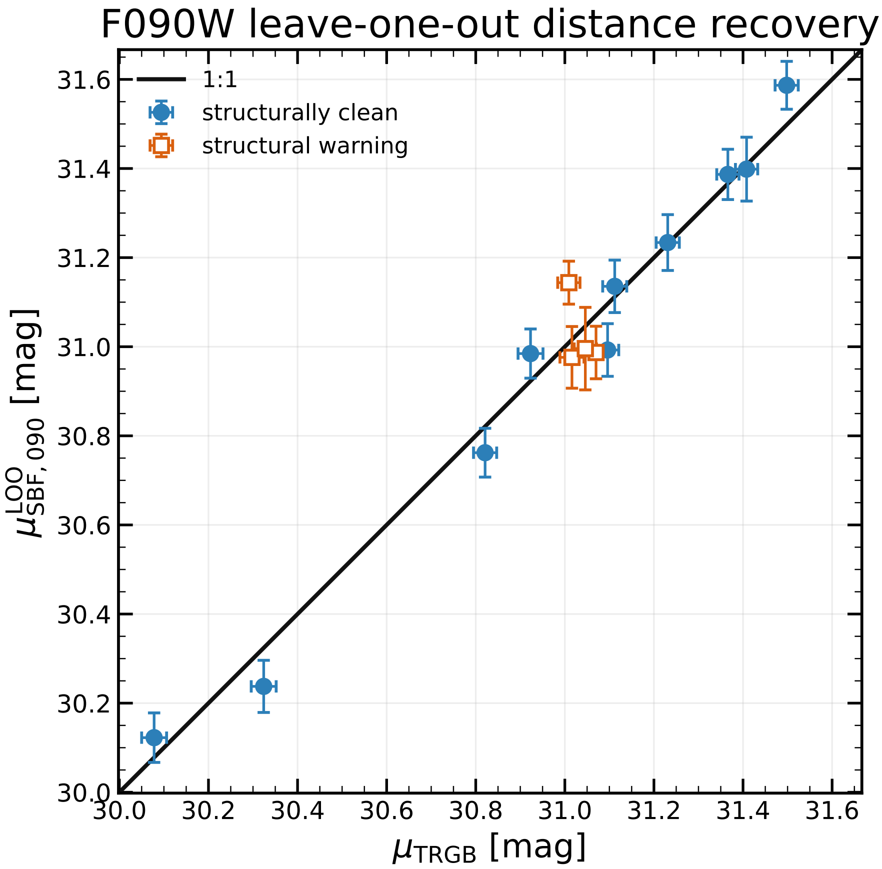

Figure 11: Color-dependent F090W LOO distance moduli compared with the individual Paper IV thin-slice TRGB moduli. Each target is excluded from its own calibration and the dashed line is equality. Error bars are internal; the common absolute zero point is not shown.

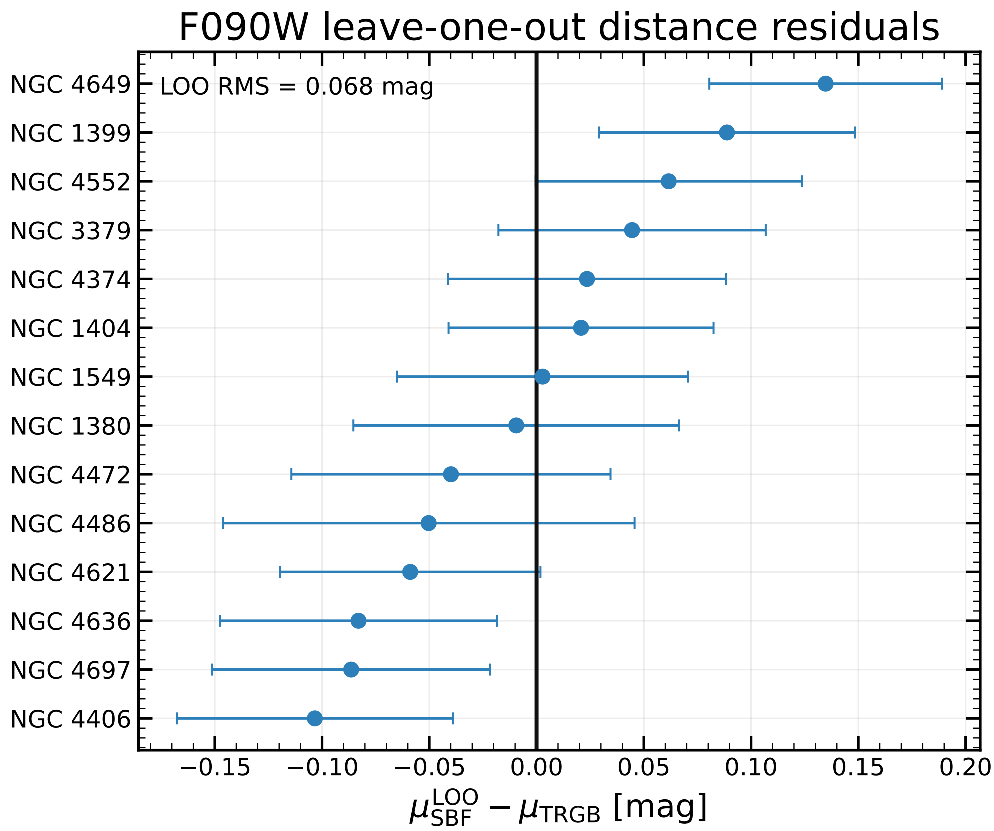

Figure 12: Residuals of the color-dependent F090W LOO predictions relative to the individual Paper IV thin-slice TRGB moduli. The vertical dashed line marks zero residual. Error bars contain the galaxy-dependent predictive terms but exclude the shared absolute TRGB zero-point scale.

Table 8: Primary F090W comparison. Errors on a and b are galaxy-level bootstrap standard deviations. The median distance uncertainty in the last column is a full absolute 1σ value including the common 0.047-mag TRGB scale.

Model	a (mag)	b	$\sigma_{\rm int}$ (mag)	AICc	LOO RMS (mag)	Median σ_D (Mpc)
Constant	-1.949 ± 0.024	0.00 ± 0.00	0.086	-21.84	0.099	0.81
Linear	-1.962 ± 0.016	1.89 ± 0.44	0.049	-30.98	0.068	0.57

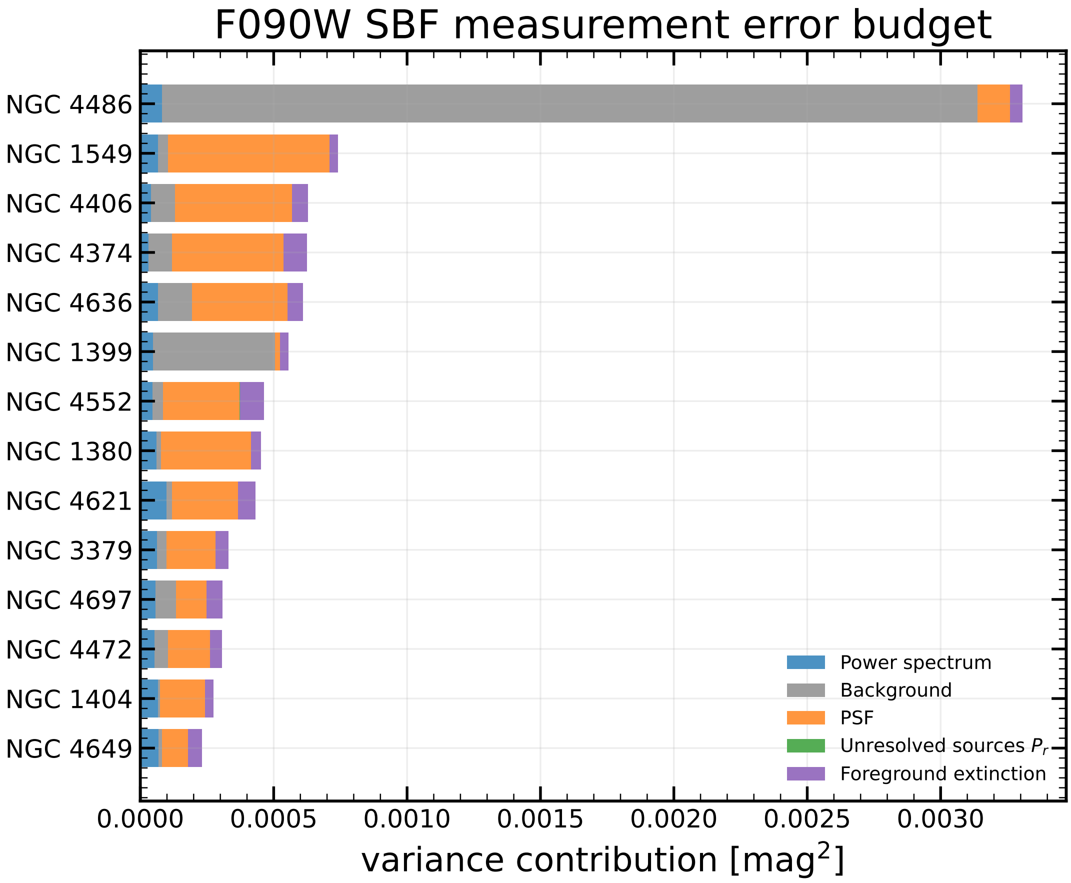

Figure 13: F090W apparent-SBF measurement budget in the same variance representation as Figure 6. Each horizontal segment is one contribution σ_i^2 and galaxies are ordered by total measurement variance.

Table 9: Uncertainty convention used to place the five distance columns on a common footing. "Individual" includes predictive scatter where applicable; "common" moves the corresponding absolute scale coherently.

Method	Included terms	Common term
TRGB Paper IV	Published individual uncertainty	Add 0.047 mag
Jensen III F110W	$\overline{m}$, color, slope, and 0.060-mag population scatter	Add 0.063 mag
Jensen 2015 F160W	Published final combined uncertainty	Already included
This work F150W	Measurement, intrinsic scatter, and finite LOO calibration	Add 0.047 mag
This work F090W	Measurement, color, intrinsic scatter, and finite LOO calibration	Add 0.047 mag

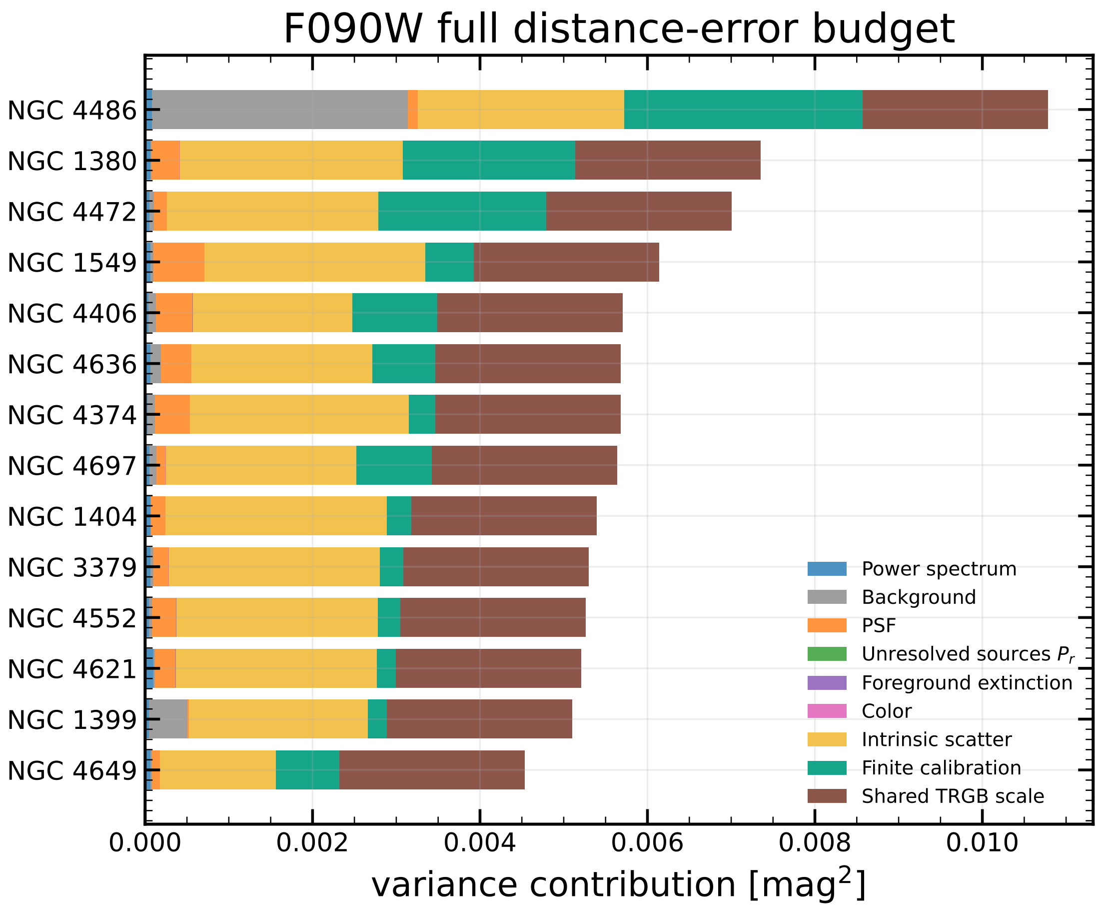

Figure 14: Full F090W LOO distance-error budget in the same variance representation as Figure 7. The color-propagation term is included for the adopted linear calibration; intrinsic scatter, finite calibration, and the coherent TRGB absolute scale are added as distinct components.

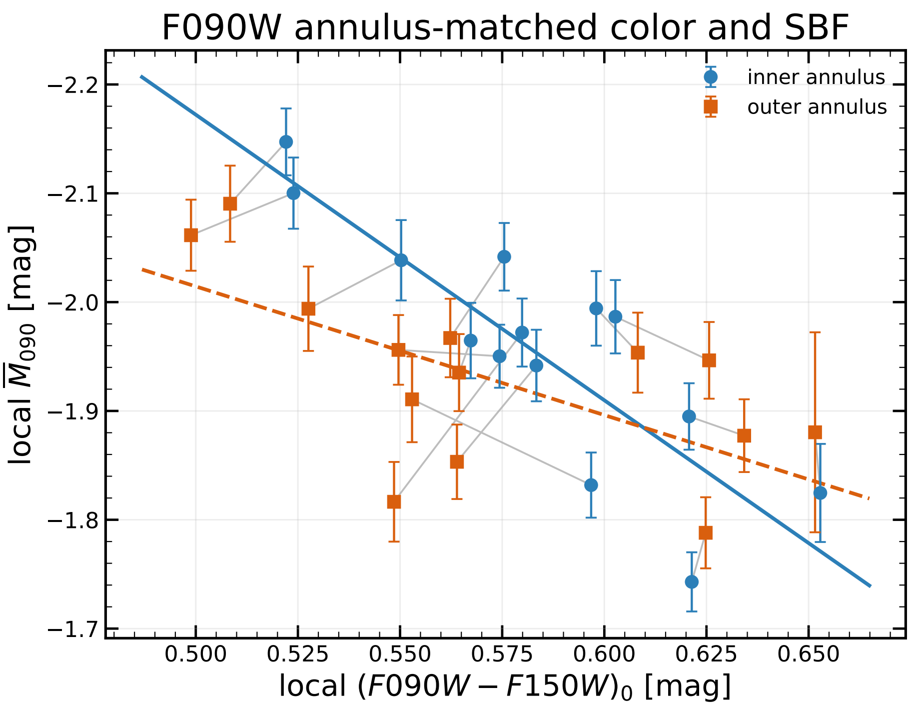

Figure 15: Annulus-matched local color and F090W SBF in the same plotting grammar as Figure 8. Blue circles and the solid blue line are the inner annuli; orange squares and the dashed orange line are the outer annuli. The rings are correlated measurements of the same galaxies and are not treated as 28 independent calibrators.

Table 10: Final distances in Mpc. TRGB IV is the individual Paper IV thin-slice anchor; Jensen III is the Paper III HST F110W reconstruction; Jensen 2015 is the separately labelled legacy HST F160W result. This work uses color-independent F150W LOO distances and color-dependent F090W LOO distances. Every printed uncertainty is the full absolute 1σ value; common terms are included but remain correlated. Ellipses mark objects absent from a comparison sample.

Galaxy	TRGB IV	Jensen III F110W	Jensen 2015 F160W	This work F150W	This work F090W
NGC 1380	19.12 ± 0.47	18.98 ± 0.81	20.12 ± 1.19	18.08 ± 0.78	19.04 ± 0.75
NGC 1399	19.93 ± 0.49	19.80 ± 0.85	20.30 ± 1.19	18.66 ± 0.81	20.77 ± 0.68
NGC 1404	18.76 ± 0.46	19.44 ± 0.83	19.84 ± 1.22	18.94 ± 0.87	18.94 ± 0.64
NGC 1549	17.63 ± 0.44	...	...	16.83 ± 0.75	17.65 ± 0.64
NGC 3379	10.37 ± 0.26	...	...	11.12 ± 0.47	10.58 ± 0.35
NGC 4374	16.69 ± 0.42	...	...	17.04 ± 0.79	16.87 ± 0.59
NGC 4406	16.57 ± 0.41	...	...	16.25 ± 0.75	15.80 ± 0.55
NGC 4472	15.97 ± 0.40	16.11 ± 0.69	16.19 ± 0.98	15.23 ± 0.69	15.68 ± 0.60
NGC 4486	16.19 ± 0.40	...	...	15.95 ± 0.81	15.82 ± 0.76
NGC 4552	15.30 ± 0.39	15.70 ± 0.67	...	16.36 ± 0.69	15.74 ± 0.53
NGC 4621	14.59 ± 0.36	...	...	15.15 ± 0.68	14.20 ± 0.47
NGC 4636	16.37 ± 0.40	16.30 ± 0.70	...	16.60 ± 0.77	15.75 ± 0.55
NGC 4649	15.91 ± 0.39	16.19 ± 0.71	15.86 ± 0.98	16.49 ± 0.75	16.93 ± 0.53
NGC 4697	11.61 ± 0.29	11.66 ± 0.50	...	11.49 ± 0.53	11.16 ± 0.39

10 Discussion

10.1 Accuracy of the method

The two passbands give different, internally coherent population behavior. F150W reaches 0.0925-mag LOO RMS (4.3 per cent in distance) with one constant zero point, and adding color does not help. F090W requires color and reaches 0.0684-mag LOO RMS (3.2 per cent); forcing it to be constant degrades the RMS to 0.0992 mag. This is not a contradiction: SBF is weighted toward the brightest evolved stars, and its population response changes with passband. The empirical statement is limited to the present color range and 14 calibrators; it is not a universal claim that F150W can never depend on color.

The error model is broadly consistent with the realized residuals: the F150W and F090W validation-pull RMS values are 1.08 and 1.12. NGC 4649 is the sole F090W residual beyond 2σ. After all methods are placed on a full absolute-error convention, the median Paper III F110W and present F150W uncertainties are similar, while F090W is smaller. This comparison also shows why raw P_0 fit precision alone is not distance precision: population scatter and finite calibration dominate after the image has been measured.

None of these LOO numbers is an external validation. Every omitted galaxy is still predicted from the same instrument, reduction, PSF construction, and TRGB scale. An independent galaxy sample is required to test external accuracy and common systematic offsets.

10.2 Remaining limitations

Several measurement systematics remain physically important. First, i2d noise is correlated. The attempted empirical $N(k)$ component is too degenerate with $E(k)$ over the fitted interval to stabilize a third coefficient, so the failure of that extension is documented rather than hidden behind a more complicated fit. Second, the retained 3.5σ winsorization is applied only after normalization, with one object-specific threshold estimated from the complete valid positive-model support. The all-galaxy branch comparison shows that changing the order relative to the former raw-residual prescription is measurable, even though the synthetic branch-recovery test finds no additional bias for an injected Gaussian field. The selected threshold and ordering are therefore fixed methodological choices, not negligible corrections. Third, the empirical PSF comparison is not closed by a confirmed isolated star. The 129–257-pixel model comparison does, however, bound the current finite-stamp shape effect. Compact-source completeness and the P_r model remain approximate; the explicit 0–$2P_r$ test bounds their impact on the F150W calibration without pretending that completeness itself has been measured. Finally, the F090W slope is significant in the full sample but varies between the two rings, and its formal pixel-color errors are much smaller than likely image-wide calibration systematics. The color-floor and fixed-subset tests limit this concern, but a larger independent sample is needed before fitting more flexible population laws.

11 Conclusions

1. All 14 GO-3055 galaxies are retained in the primary analysis with individual Paper IV thin-slice TRGB anchors and no manual exclusions.
2. The adopted F150W calibration is $\overline{M}_{150,0} = -3.208 \pm 0.024_{\mathrm{stat}} \pm 0.051_{\mathrm{common}}$ mag. It recovers the TRGB distances with 0.0925 mag LOO RMS (4.3 per cent) and negligible mean bias.
3. F150W does not require a color dependence: a linear color law worsens the LOO RMS to 0.0981 mag, its AICc is worse by 3.15, and its slope interval includes zero. It is therefore retained only as a diagnostic and is not used for F150W distances.
4. The adopted F090W calibration has slope 1.889 ± 0.442 and intrinsic scatter 0.0489 mag. It improves the LOO RMS from 0.0992 mag for a constant to 0.0684 mag (3.2 per cent), so color is used for F090W distances.
5. Median full absolute uncertainties are 0.0984 mag for F150W and 0.0752 mag for F090W. Their validation-pull RMS values are 1.08 and 1.12; NGC 4649 is the only F090W point beyond 2σ.
6. Paper IV thick-slice distances and Paper III cluster means do not materially change the recovered accuracy. Thin slices remain primary only because that experiment was pre-specified, not because they give a better answer.
7. The Virgo–Fornax step, quality cuts, and NGC 1399 exclusion remain diagnostics; none is promoted to a detected effect or used to define the main sample.
8. Jensen III's full F110W errors have median 0.70 Mpc, compared with 0.75 Mpc for F150W and 0.57 Mpc for F090W. The previously smaller Jensen error bars resulted from comparing formal and full predictive uncertainties.
9. The P_r and synthetic FFT tests are negligible at current precision. The $N(k)$ extension is rejected because it is nearly degenerate with $E(k)$; the 129-pixel PSF choice and 3.5σ winsorization remain explicit tested methodological choices.
10. Both calibrations are internal LOO cross-validations. A confirmed empirical stellar PSF and an independent galaxy sample are required before claiming external accuracy.

12 Data and code availability

The calibrated JWST observations are public through MAST under GO-3055; the archive collection is accessible through DOI 10.17909/z9ch-wk24. No proprietary data are used. The machine-readable derived products required to reproduce the tables and figures are listed in Appendix B and retained with this working paper.

The measurement and analysis code is presently the version-controlled project copy described in Appendix C.

A public archival release and permanent code DOI have not yet been assigned and must be added before submission. This statement does not claim that an unarchived local repository is publicly available.

A Measurement-pipeline diagnostic plots

The following plots document numerical sensitivity tests requested during pipeline validation. They support the systematic-error bounds quoted in the main text and do not enter either adopted distance estimator as additional data. Band-dependent tests are shown in matched F150W/F090W pairs; synthetic tests of the common Fourier algorithm are shown once.

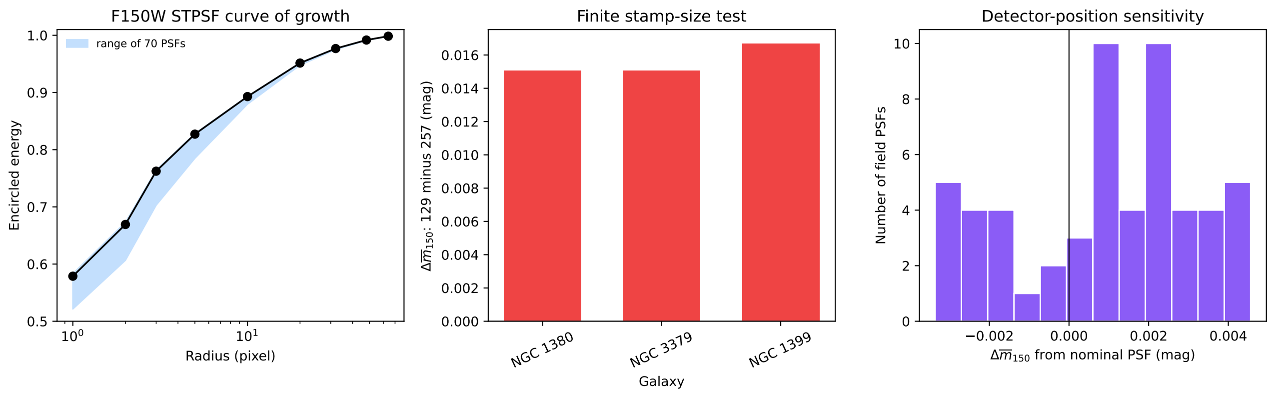

Figure 16: F150W PSF checks from the final products. Left: encircled-energy curves for the 70 saved STPSF realizations. Middle: the fitted $\overline{m}_{150}$ shift between 129- and 257-pixel stamps for the three available test galaxies. Right: shifts produced by the four detector-position variants for every galaxy. The black vertical line marks zero shift.

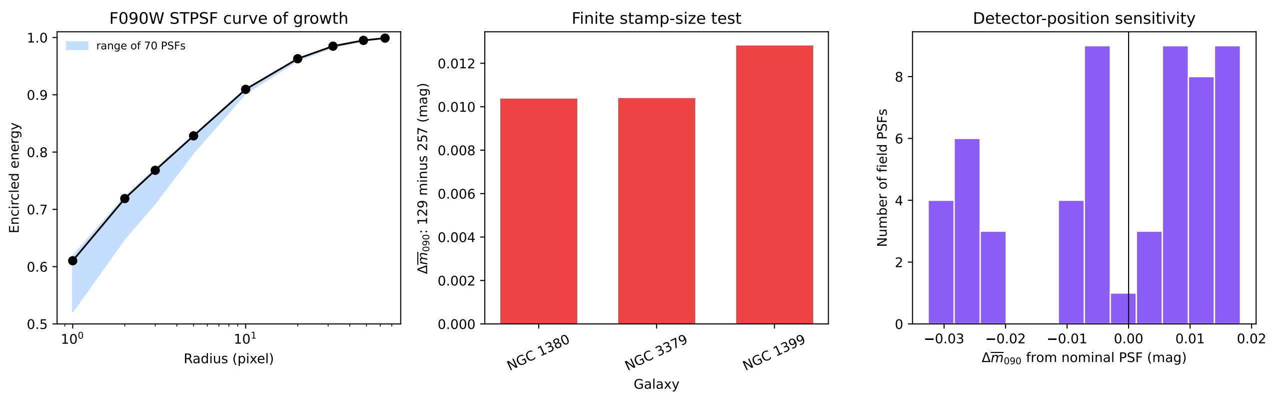

Figure 17: Direct F090W counterpart of Figure 16, with the same encircled-energy, 129-versus-257-pixel, and detector-position tests. The panels use the same final annular weighting and fitted frequency range.

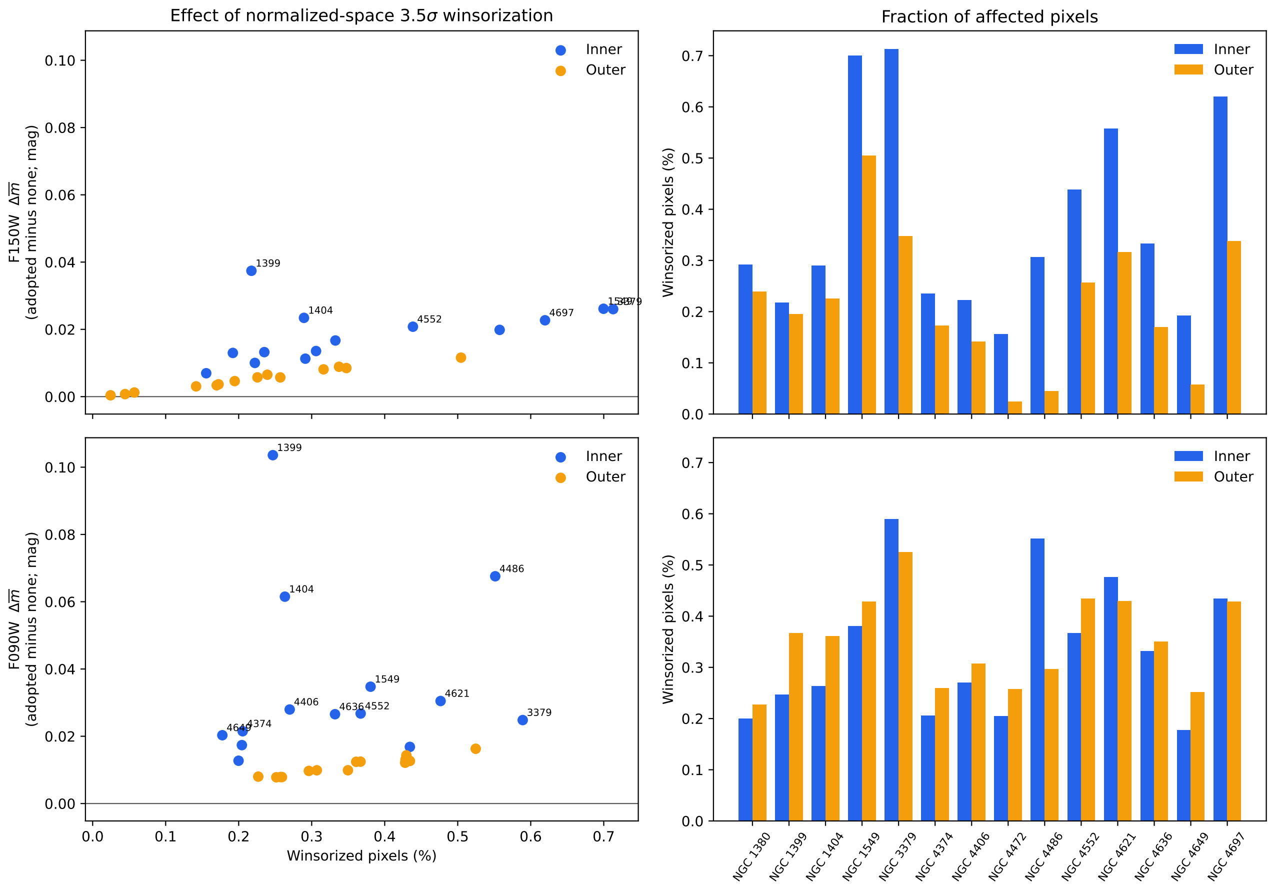

Figure 18: Matched all-galaxy test of the adopted normalized-space 3.5σ winsorization. The upper row is F150W and the lower row is F090W. Left panels show the change relative to no winsorization against the affected pixel fraction; right panels show those fractions separately for the inner and outer annuli. All values come from the final full-support products.

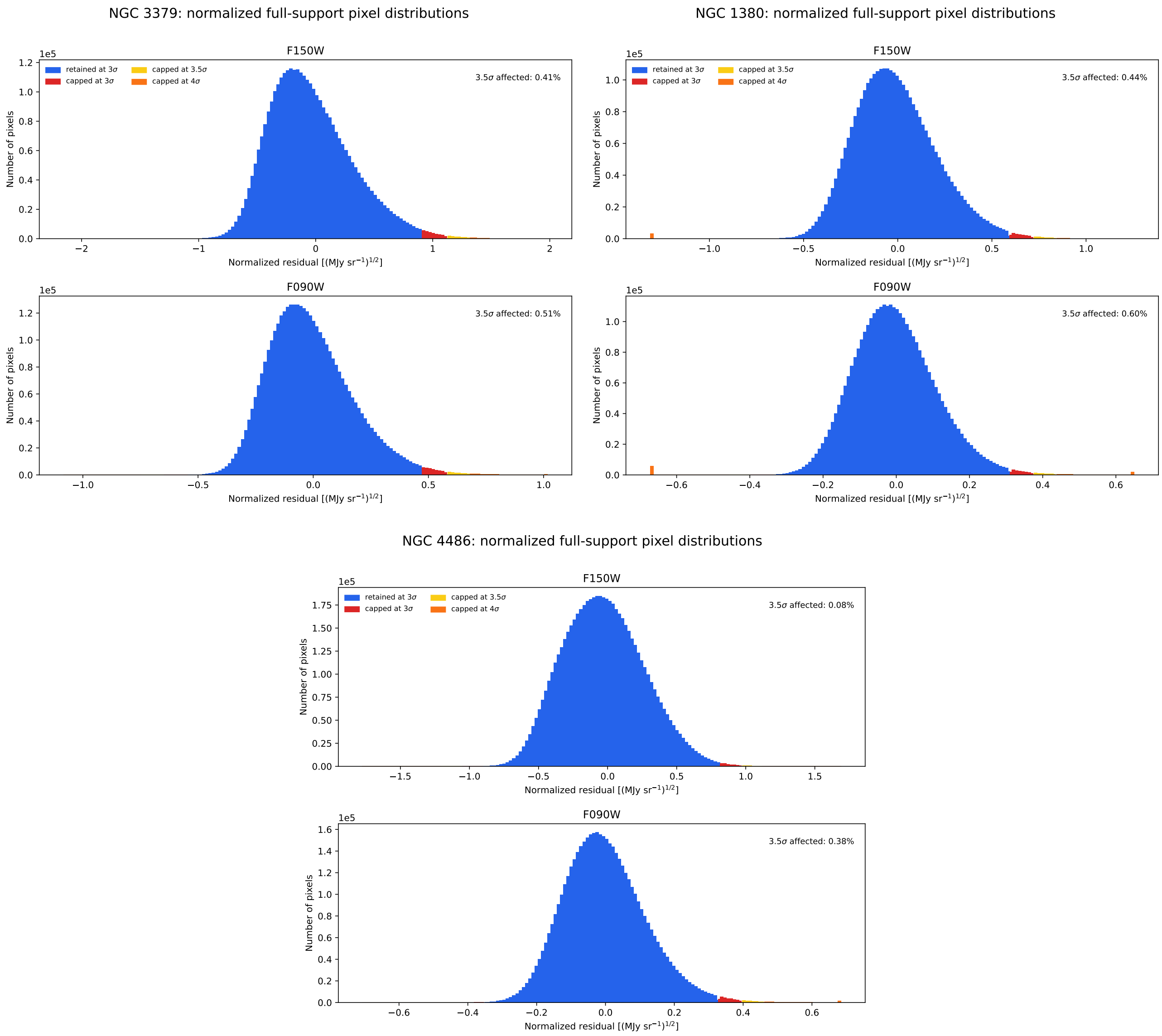

Figure 19: Final normalized full-support pixel distributions for three representative galaxies. Each target is shown in F150W and F090W. Blue is the central region retained by a 3σ rule; red, yellow, and orange mark the successive 3σ, 3.5σ, and 4σ tails. The adopted 3.5σ threshold affects less than one per cent of valid pixels in these examples.

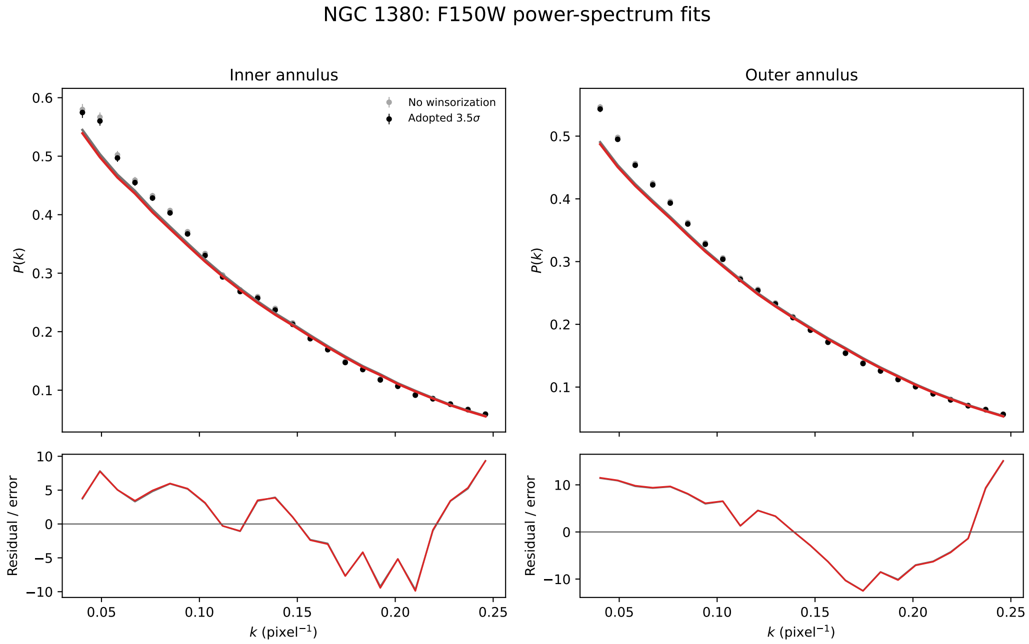

Figure 20: F150W power-spectrum fit for NGC 1380 in the two adopted annuli. Black points and red curves show the final normalized-space 3.5σ branch; grey points and curves show the otherwise identical no-winsorization branch. Points are radial-bin means of the measured two-dimensional Fourier power, and the lower panels show residuals divided by their bin errors.

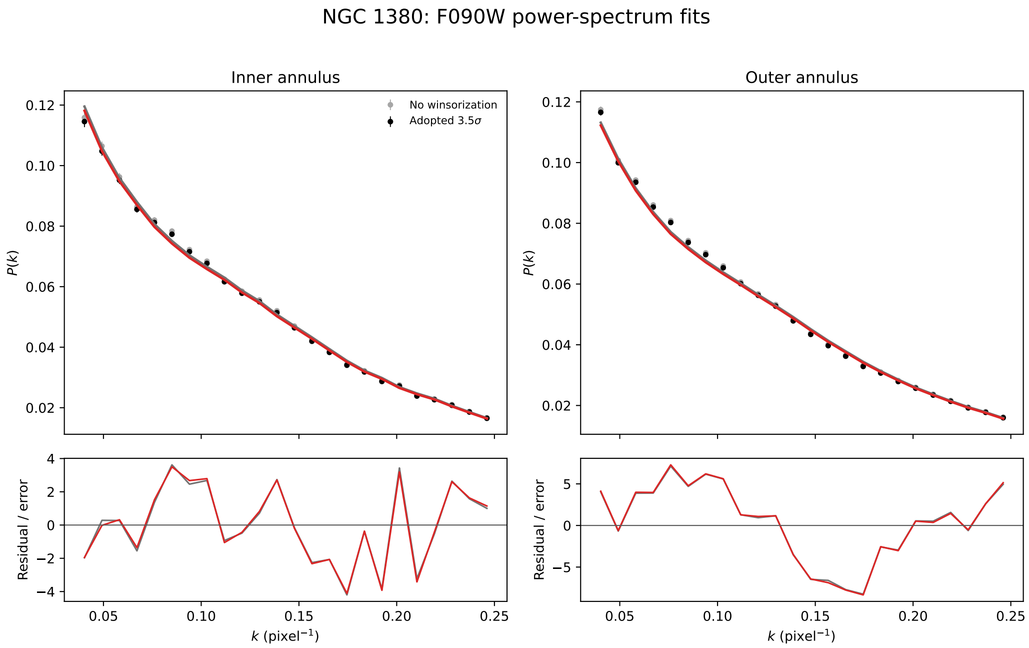

Figure 21: Direct F090W counterpart of Figure 20, using the same galaxy, annuli, frequency interval, spectral model, and winsorization comparison.

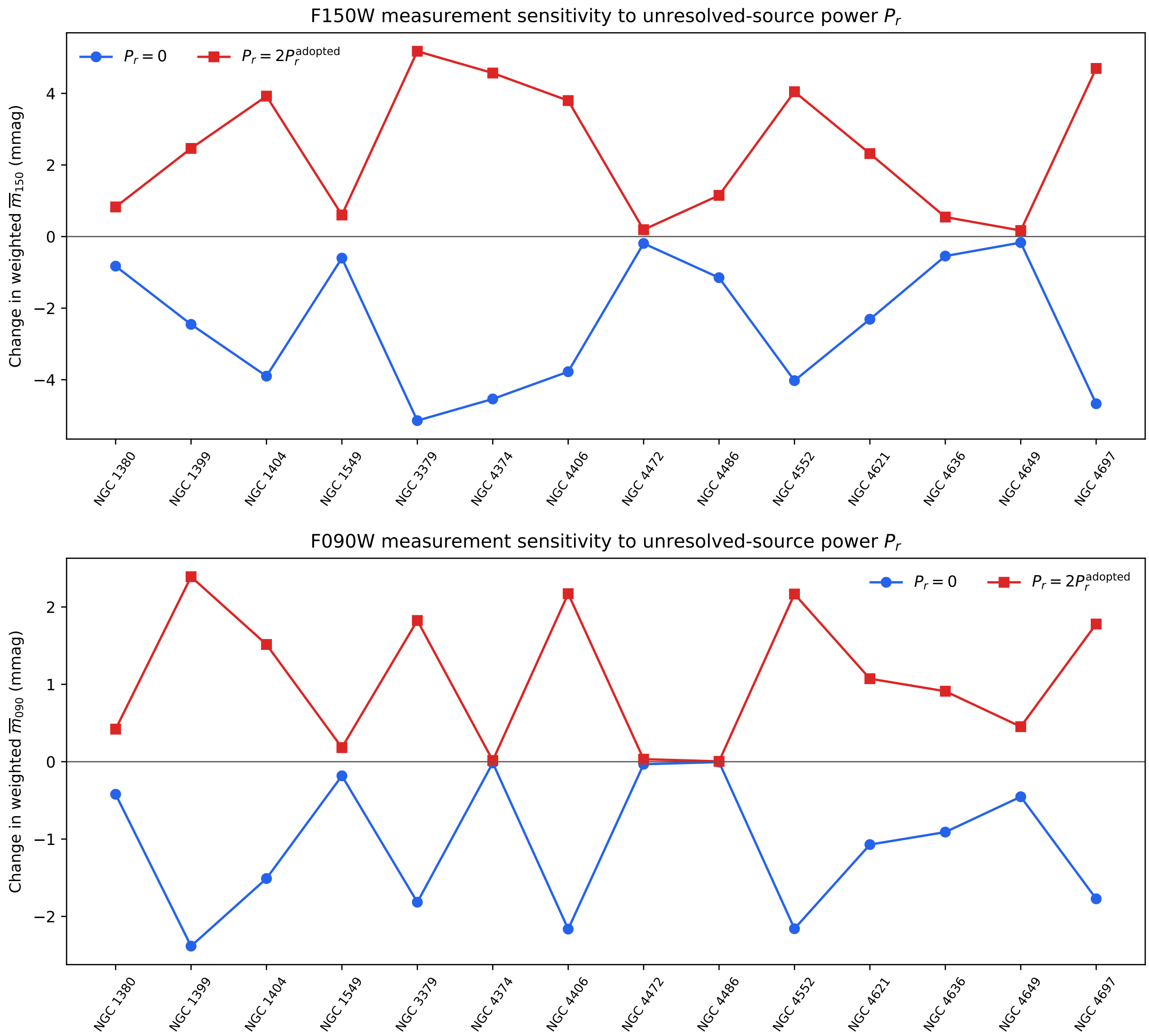

Figure 22: Direct measurement-level sensitivity to the unresolved-source correction P_r. The upper panel is F150W and the lower panel is F090W. Each curve gives the change in the inverse-variance-weighted annular fluctuation magnitude when the adopted correction is replaced by $P_r = 0$ or $2P_r$. The vertical scale is millimagnitudes.

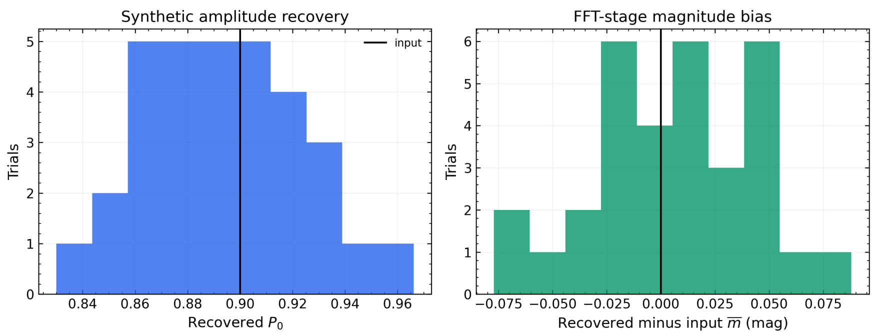

Figure 23: Recovery of known P_0 and SBF magnitude in the synthetic FFT-stage test. A representative F150W PSF is used to test the common Fourier normalization and unit conversion; this is not a separate estimate of an F150W-only systematic.

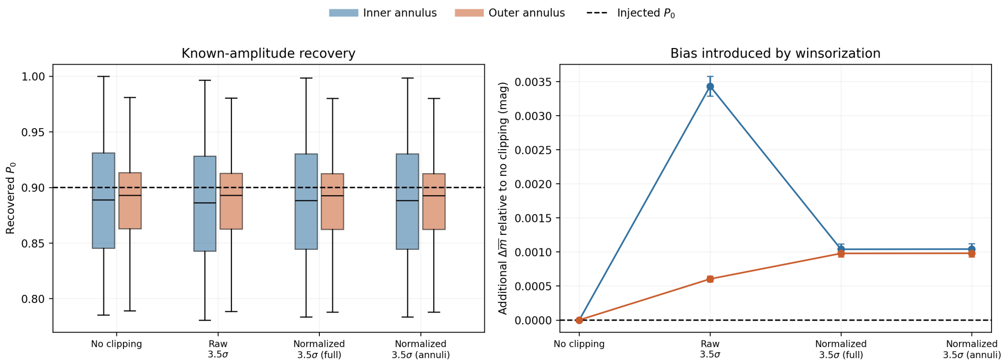

Figure 24: Synthetic recovery test for the order of normalization and winsorization. Each spectral branch is applied to the same injected SBF and white-noise realization before the inner and outer annuli are fitted. The comparison isolates this ordering only; it is not an end-to-end test of sky, isophotal, catalogue, or correlated-noise systematics. In the left panel the line inside each box is the median, the box spans the interquartile range, and the whiskers extend to the most extreme values within 1.5 interquartile ranges; these elements do not denote a 68 per cent interval.

B Auxiliary scientific and quality diagnostics

The following figures and table retain useful checks that do not define the primary calibration. They are separated from the main Results section so that the central claims are not obscured by exploratory host-luminosity and small-overlap comparisons.

For F150W, the aperture-magnitude test detects no direct correlation for all 14 objects (Pearson $p = 0.089$), Virgo alone ($p = 0.452$), or the three Fornax objects ($p = 0.954$). After subtracting the rejected F150W linear color relation, the corresponding diagnostic values are 0.053, 0.321, and 0.900. F090W shows an apparent raw relation for all objects ($p = 0.021$) and Virgo alone ($p = 0.0047$), but it disappears after subtracting the adopted F090W color calibration ($p = 0.209$ and 0.243, respectively); the three-object Fornax values are 0.074 and 0.186. Thus the F090W trend does not provide an independent host-luminosity dependence. In either band, the fixed angular aperture is also confounded with environment and distance.

B.1 Cross-band calibration diagnostics

B.2 Machine-readable products

The final TRGB/F110W/F160W/F150W/F090W distance table is `runs/sbf2_go3055/analysis/tables/go3055_final_distance_comparison.csv`. Its long-form decomposition into central, individual, common, and total uncertainties in both magnitudes and Mpc is `runs/sbf2_go3055/analysis/tables/go3055_final_distances_error_components.csv`. The predictive errors are in `runs/sbf2_go3055/analysis/tables/go3055_distance_predictions_full_errors.csv`; the component table is `runs/sbf2_go3055/analysis/tables/go3055_error_budget.csv`; anchor tests are in `runs/sbf2_go3055/analysis/tables/go3055_distance_anchor_sensitivity.csv`.

Additional machine-readable systematics products are `runs/sbf2_go3055/analysis/tables/go3055_psf_129_vs_257_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_psf_detector_position_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_Pr_calibration_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_synthetic_fft_summary.csv`, and `runs/sbf2_go3055/analysis/tables/go3055_structural_quality_and_physical_annuli.csv`. The F090W measurements, model comparison, LOO distances, and error budget are `runs/sbf_f090w_go3055/analysis/tables/go3055_f090w_master.csv`, `runs/sbf_f090w_go3055/analysis/tables/go3055_f090w_color_model_comparison.csv`, `runs/sbf_f090w_go3055/analysis/tables/go3055_f090w_distances_all_models.csv`, and `runs/sbf_f090w_go3055/analysis/tables/go3055_f090w_distance_error_budget.csv`. The consolidated three-galaxy noise, winsorization, and PSF experiment is `runs/sbf2_systematics/pilot/tables/sbf2_three_galaxy_systematics_pilot.csv`; its adopted decisions are recorded separately in `runs/sbf2_systematics/pilot/tables/sbf2_three_galaxy_systematics_decision.csv`. The final all-14 normalization/winsorization comparison is retained under `runs/sbf2_normalized_winsor/batch/aggregates/`; its per-target status table is `runs/sbf2_normalized_winsor/batch/target_status.csv`, and the synthetic branch-recovery summaries are in `runs/sbf2_normalized_winsor/recovery/`. The compact uncertainty summaries printed in the article are also available as `runs/sbf2_go3055/analysis/tables/go3055_individual_error_summary.csv`, `runs/sbf2_go3055/analysis/tables/go3055_predictive_error_summary.csv`, and `runs/sbf2_go3055/analysis/tables/go3055_common_systematics.csv`.

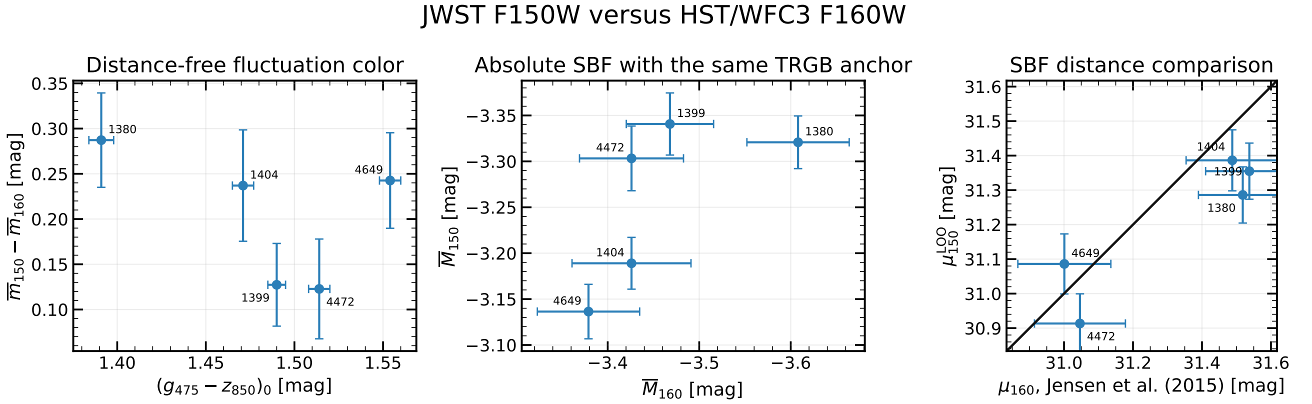

Figure 25: Comparison with the five independently reduced Jensen et al. (2015) F160W overlaps for the F150W measurements. From left to right: the distance-free fluctuation color, absolute SBF magnitudes using the same TRGB anchor, and LOO SBF distance moduli. The black line in the right panel is equality. The five-object sample is retained as a cross-band consistency check and does not define a new calibration.

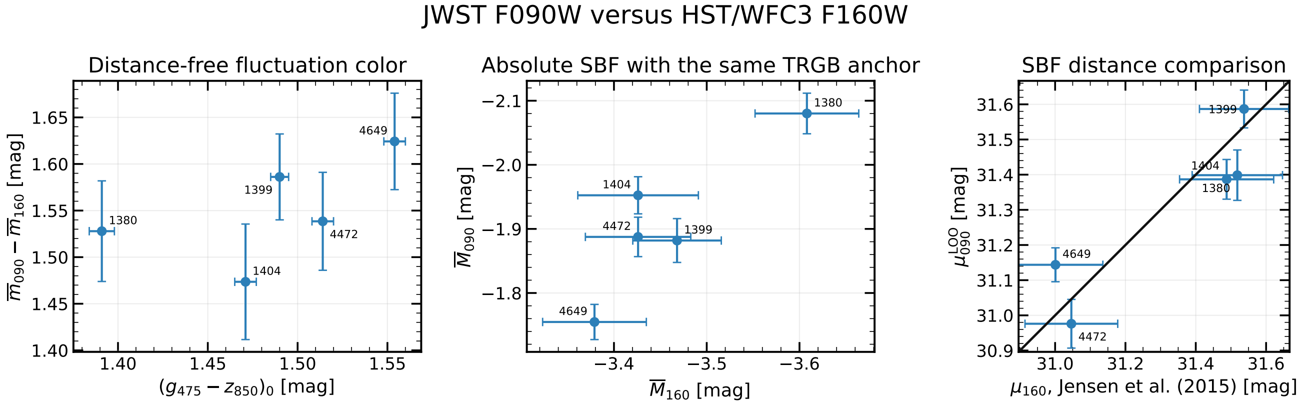

Figure 26: The direct F090W counterpart of Figure 25, using the same five Jensen et al. (2015) F160W overlaps and the same three comparisons. This diagnostic likewise does not enter the adopted F090W calibration.

Table 11: Predeclared structural-quality diagnostics. The automatic flag is set at a structure score of 0.80; the visual score is on the user-defined 0–10 scale. These flags define sensitivity samples and do not remove any galaxy from the primary 14-object calibration.

Galaxy	Visual score	Auto score	Auto flag	Coverage flag	Predeclared issue
NGC 1380	7.00	0.40	No	Yes	incomplete outer model coverage
NGC 1399	9.50	0.05	No	No	
NGC 1404	8.00	0.12	No	No	
NGC 1549	9.00	0.18	No	No	
NGC 3379	10.00	0.26	No	No	
NGC 4374	8.00	0.15	No	No	
NGC 4406	8.00	0.64	No	No	
NGC 4472	4.00	1.61	Yes	No	strong large-scale structure
NGC 4486	3.00	1.47	Yes	No	jet residual; strong large-scale structure
NGC 4552	9.00	0.20	No	No	
NGC 4621	7.00	0.35	No	No	
NGC 4636	6.00	0.97	Yes	No	strong large-scale structure
NGC 4649	5.00	1.40	Yes	No	strong large-scale structure
NGC 4697	6.00	0.40	No	Yes	incomplete outer model coverage

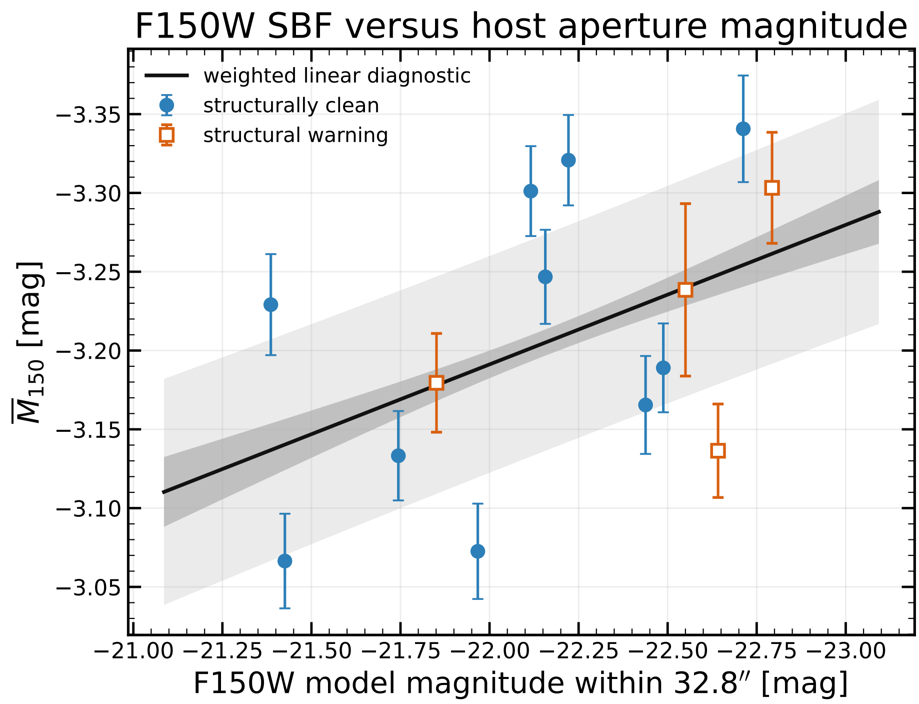

Figure 27: F150W SBF versus the smooth-model magnitude inside the fixed 32.8-arcsec aperture. This is an aperture luminosity, not a total galaxy magnitude, and the same angular radius covers different physical radii. The black line is a weighted exploratory regression; dark and light grey regions show its 68 per cent mean-confidence and predictive intervals. The relation does not enter the adopted distance estimator.

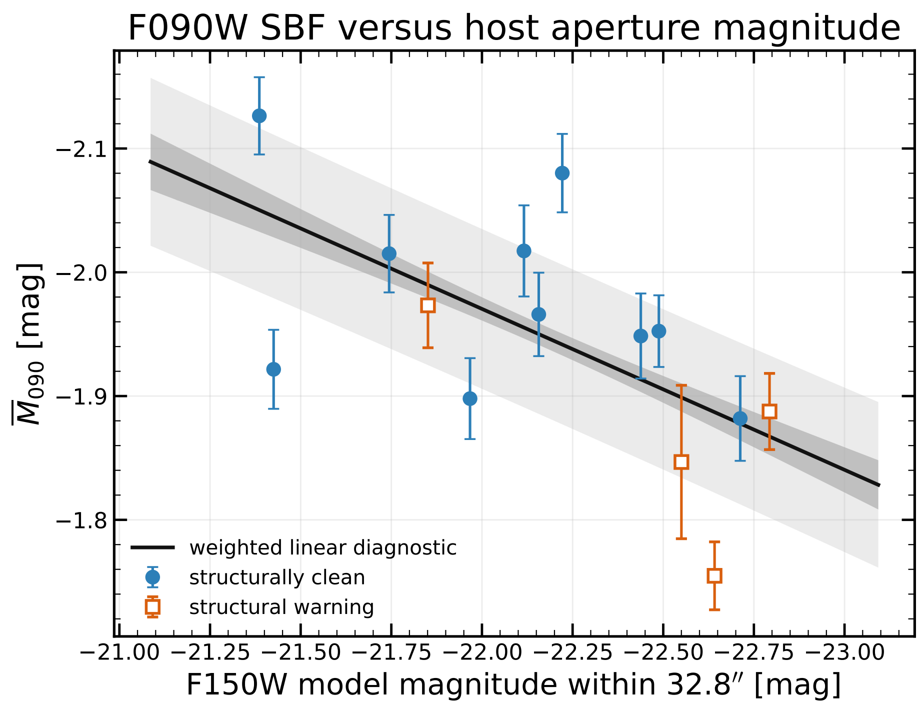

Figure 28: The direct F090W counterpart of Figure 27, using the identical F150W smooth-model aperture magnitude on the horizontal axis and the same fitting and interval conventions. This exploratory relation is not used in the F090W distance calibration.

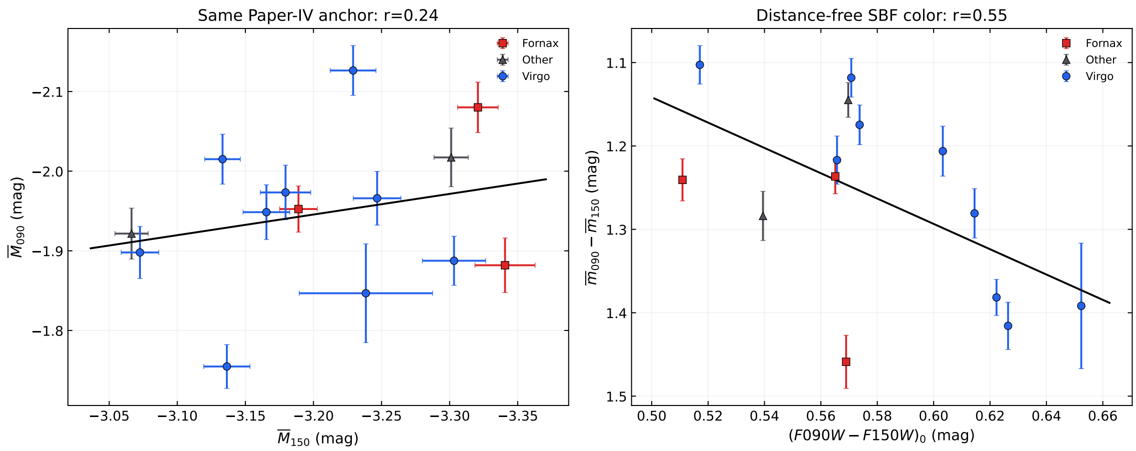

Figure 29: Direct comparison of the independently measured F090W and F150W fluctuation quantities for the same 14 galaxies. The panels are cross-band diagnostics and do not constitute an independent distance validation because both measurements share images, masks, galaxy models, and TRGB anchors.

C Reproducibility

The galaxy modelling and source-mask notebook is `code/sbf-2.ipynb`; its dedicated GO-3055 runner is `code/run_sbf_2_batch.py`. The final normalized-full spectral branch is implemented in `code/sbf2_normalized_winsor_core.py` and executed sequentially by `code/run_sbf_2_normalized_winsor.py`. It reuses the saved galaxy model, masks, PSFs, annular geometry, and cached $E(k)$ rather than repeating isophotal modelling. Exploratory fits and diagnostic figures are generated and displayed inline by `code/sbf-2-graph.ipynb`, which reads completed batch products and does not repeat the SBF measurement. The three-galaxy winsorization, noise, and 129-versus-257-pixel PSF diagnostics are generated by `code/sbf-2-systematics.ipynb`. The all-galaxy spectral-branch comparison is summarized in `code/sbf-2-normalized-winsor.ipynb`; its synthetic recovery is implemented by `code/sbf2_normalized_winsor_recovery.py`. The adopted constant-calibration tables, distance-recovery figures, error-budget figure, comparison tables, and provenance records are generated from those saved products by `code/build_sbf2_article_tables.py`. The F090W measurement is run by `code/run_sbf_f090w.py`; its calibration, model comparisons, LOO distances, and inline figures are generated by `code/sbf-f090w-graph.ipynb`, whose reproducible source builder is `code/build_sbf_f090w_graph_notebook.py`.

The implementation uses NumPy, SciPy, Astropy, Photutils, Matplotlib, and STPSF [17–23]. Exact installed versions, rather than only literature citations, are given in Table 13.

The 28 input i2d headers are heterogeneous archive products rather than a uniform local reprocessing: 2 files record JWST pipeline 1.19.1, CRDS 12.1.11, and `jwst_1413.pmap`; 22 record pipeline 1.20.2, CRDS 13.0.6, and `jwst_1464.pmap`; and 4 F090W files record pipeline 2.0.1, CRDS 13.1.13, and `jwst_1535.pmap`. Exact counts are generated directly from the files used by the batch and retained in `go3055_input_calibration_provenance.csv`. This records the reduction state exactly, but does not prove that mixing contexts is harmless; uniform reprocessing is a residual reproducibility/systematics check. The base model-and-mask run records Git HEAD `5d0fd1496474...` and the exact executed `sbf-2.ipynb` SHA256 `a1828d2d8d97...`. That snapshot defines the galaxy model, masks, annuli, PSFs, and reusable $E(k)$ caches, but does not by itself define the adopted spectrum measurement. The latter is fixed jointly by the hashes of `sbf2_normalized_winsor_core.py` and `run_sbf_2_normalized_winsor.py`, the version/branch pair `sbf2-normalized-winsor-v3/normalized_full_3p5`, and the per-target experiment keys in the v3 result JSON files. The frozen F150W science-result commit is `d89e09482978...`. Full identifiers are retained in `go3055_software_provenance.csv`.

With the virtual environment active, the reproducible analysis commands are run from `code/`:

```
py run_sbf_2_batch.py \
  --no-download --no-cleanup-inputs
py run_sbf_2_normalized_winsor.py --stop-on-error
py sbf2_normalized_winsor_recovery.py
py run_sbf_f090w.py
jupyter nbconvert \
  --to notebook --execute --inplace \
  sbf-2-graph.ipynb \
  --ExecutePreprocessor.timeout=7200
jupyter nbconvert \
  --to notebook --execute --inplace \
  sbf-f090w-graph.ipynb \
  --ExecutePreprocessor.timeout=7200
py build_sbf2_article_tables.py
cd ../texts/paper_work
latexmk -pdf \
  -outdir=build -interaction=nonstopmode \
  -halt-on-error go3055_jwst_sbf_article_draft.tex
```

The notebooks save every publication figure and display the same figure inline during interactive execution.

Table 12: Fixed processing configuration for the reported run. Values not listed here remain recorded in the executed notebook snapshot.

Stage	Adopted configuration
Input and bands	JWST/NIRCam i2d; independent F090W and F150W SBF signals; F090W–F150W color; native pixel scale $\simeq 0.03123$ arcsec
Science regions	circular 8.2–16.4 and 16.4–32.8 arcsec annuli
Isophotes	initial semimajor axis 50 pixels; free ellipticity and position angle; 10-pixel sampling; maximum 2000 pixels; real-pixel fit with filled-gap fallback
Compact masks	primary 3.5σ, 9 connected pixels, 2-pixel dilation; residual catalogue 2.5σ, 4 connected pixels, deblending, 3-pixel dilation
Residual limiting	divide by $\sqrt{M}$ first; estimate one symmetric 3.5σ interval from the complete valid positive-model support; winsorize and save the full normalized FITS; select each ring and subtract its mean only immediately before the FFT. Raw, no-winsorization, and annulus-union branches are diagnostics only
PSF	STPSF; nearest time-matched WSS OPD; nominal plus four detector offsets; 129 pixels; 7 wavelengths; unit normalization
Fourier model	80 radial bins; 64 Monte Carlo realizations per $E(k)$ and PSF; $P_0E(k) + P_1$; primary $k = 0.04$–0.25; $k_{\min} = 0.01, 0.03$ sensitivity windows
Unresolved sources	Gaussian GCLF plus background-galaxy power law; 12-mag faint integration; 25 per cent P_r uncertainty; 0, P_r, and $2P_r$ tests
Calibration	all 14 primary; individual Paper IV thin-slice anchors; F150W constant and F090W linear adopted relations; fitted intrinsic scatter; 300 LOO prediction bootstraps; galaxy-level coefficient bootstraps. Nonlinear relations are diagnostic only

Table 13: Software and provenance identifiers. Long hashes are abbreviated in print and retained in full in the accompanying CSV.

Item	Version or identifier
Input i2d products (2/28)	pipeline 1.19.1; CRDS 12.1.11; jwst_1413.pmap
Input i2d products (22/28)	pipeline 1.20.2; CRDS 13.0.6; jwst_1464.pmap
Input i2d products (4/28)	pipeline 2.0.1; CRDS 13.1.13; jwst_1535.pmap
Python	3.13.2
numpy	2.3.5
astropy	7.2.0
photutils	2.3.0
stpsf	2.1.0
scipy	1.16.3
matplotlib	3.10.8
pandas	2.3.3
Adopted F150W SBF contract	sbf2-normalized-winsor-v3; normalized_full_3p5; kmin=0.04
Normalized SBF core SHA256	`33f9e794e329...`
Normalized SBF runner SHA256	`2505a6de87dc...`
Product-generation Git HEAD	`5d0fd1496474...`
Executed sbf-2.ipynb SHA256	`a1828d2d8d97...`
Current sbf-2-graph.ipynb SHA256	`f21849efcbb3...`
Current sbf-2-systematics.ipynb SHA256	`32cfe6de01f8...`
Current sbf-f090w-graph.ipynb SHA256	`94b312a69979...`
F090W runner SHA256	`af641dd553cf...`
F090W graph builder SHA256	`a4e965b84ecd...`
Article-table builder SHA256	`7af50143cb81...`
F150W science-freeze Git commit	`d89e09482978...`

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